Math 110 Midterm 2 Review

Notes Ch. 8–13, HW4–7

§1 Review

§1.1 Background from midterm 1

Null space, range, rank–nullity. A linear map from a space to itself is an **operator**, and $\mathcal{L}(V) = \mathcal{L}(V, V)$. For $T \in \mathcal{L}(V, W)$,

$$\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T.$$

T is injective iff $\operatorname{null} T = \{0\}$. For an *operator* on a finite-dimensional space, rank–nullity gives:

$$T \text{ injective} \iff T \text{ surjective} \iff T \text{ invertible}.$$

This is used constantly below. For example, "$T - \lambda I$ is not injective" and "$T - \lambda I$ is not invertible" mean the same thing.

Direct sums. $V = U \oplus W$ means every $v \in V$ can be written as $v = u + w$ with $u \in U$, $w \in W$, in exactly one way. To check it, show either

- $U + W = V$ and $U \cap W = \{0\}$, or
- $U \cap W = \{0\}$ and $\dim U + \dim W = \dim V$, or
- $U + W = V$ and $\dim U + \dim W = \dim V$

(the last two work because $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$). For several subspaces, $V = U_1 \oplus \cdots \oplus U_m$ means each v is $u_1 + \cdots + u_m$ ($u_i \in U_i$) in exactly one way. Then putting together a basis of each U_i gives a basis of V, and $\dim V = \dim U_1 + \cdots + \dim U_m$. To check that a sum $U_1 + \cdots + U_m$ is direct, it is enough to show that $0 = u_1 + \cdots + u_m$ with $u_i \in U_i$ forces every $u_i = 0$.

Matrices of linear maps. If $v_1, \ldots, v_n$ is a basis of V and $w_1, \ldots, w_m$ a basis of W, then column j of $\mathcal{M}(T)$ is the list of coefficients of Tv_j in the basis $w_1, \ldots, w_m$. Change of basis: let A be the matrix of an operator in the standard basis of $\mathbb{F}^n$, and let the columns of an invertible matrix S be another basis $s_1, \ldots, s_n$. Then the matrix of the same operator in the basis $s_1, \ldots, s_n$ is $S^{-1}AS$. Two matrices A and $B = S^{-1}AS$ are called **similar**: they describe the same operator in two different bases.

Polynomials. These words are used constantly from Section 1.11 on.

- The **degree** of $p(x) = a_m x^m + \cdots + a_1 x + a_0$ with $a_m \neq 0$ is m; a_m is the **leading coefficient**. p is **monic** if its leading coefficient is 1.
- λ is a **root** (or **zero**) of p if $p(\lambda) = 0$.
- q **divides** p if $p = qr$ for some polynomial r. Example: $x - 1$ divides $x^2 - 1 = (x - 1)(x + 1)$.
- **Division with remainder**: for $q \neq 0$, every p can be written $p = qr + s$ with $s = 0$ or $\deg s < \deg q$ (long division). In particular λ is a root of p iff $(x - \lambda)$ divides p.
- p **splits** over $\mathbb{F}$ if it is a product of degree-1 factors with roots in $\mathbb{F}$: $p = c(x - \lambda_1) \cdots (x - \lambda_m)$, $\lambda_i \in \mathbb{F}$. A root is **repeated** if it appears more than once in this list.
- A nonzero polynomial of degree m has at most m roots (each root λ splits off a factor $x - \lambda$, lowering the degree by one).
- **Fundamental Theorem of Algebra (Prop. 52).** Every nonconstant polynomial with complex coefficients has a complex root. Hence every polynomial splits over $\mathbb{C}$.
- Over $\mathbb{R}$ this fails: $x^2 + 1$ has no real root. A real quadratic $x^2 + bx + c$ with no real root ($b^2 - 4c < 0$) is called **irreducible**. Every nonconstant real polynomial is a constant times a product of factors $x - \lambda$ and irreducible quadratics $x^2 + bx + c$.

- p has a repeated root (in $\mathbb{C}$) iff p and its derivative p' have a common root. Example: $x^2 - 2x + 1 = (x-1)^2$ and $2x - 2$ share the root 1.

Determinants. For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, $\det A = ad - bc$, and A is invertible iff $\det A \neq 0$ (HW4 P1).

Eigenvalues of a matrix. An eigenvalue or eigenvector of an $n \times n$ matrix A means one of the operator $x \mapsto Ax$ on $\mathbb{F}^n$ (Section 1.11).

§1.2 Products of vector spaces

The idea. A product puts several spaces side by side. An element of $V_1 \times \cdots \times V_m$ is a list $(v_1, \ldots, v_m)$ with one entry from each space, and you add and scale entry by entry. $\mathbb{R}^n = \mathbb{R} \times \cdots \times \mathbb{R}$ is the basic example.

Proposition (Prop. 34 / LADR 3.87, 3.89)

For vector spaces $V_1, \ldots, V_m$, the product $V_1 \times \cdots \times V_m = \{(v_1, \ldots, v_m) : v_i \in V_i\}$ is a vector space with

$$(v_1, \ldots, v_m) + (w_1, \ldots, w_m) = (v_1 + w_1, \ldots, v_m + w_m), \qquad \lambda(v_1, \ldots, v_m) = (\lambda v_1, \ldots, \lambda v_m).$$

Each axiom holds because it holds in each entry. The zero vector is $(0, \ldots, 0)$ and the negative of $(v_1, \ldots, v_m)$ is $(-v_1, \ldots, -v_m)$.

Example (Ex. 39, 40)

$\mathbb{R}^n$ is $\mathbb{R} \times \cdots \times \mathbb{R}$ (n factors). $\mathbb{R}^m \times \mathbb{R}^n$ is identified with $\mathbb{R}^{m+n}$ by $\big((x_1, \ldots, x_m), (y_1, \ldots, y_n)\big) \mapsto (x_1, \ldots, x_m, y_1, \ldots, y_n)$. An element of $\mathcal{P}_1(\mathbb{R}) \times \mathbb{R}^2$ is a pair such as $\big(2 + 3x,\ (1, -1)\big)$, and $\big(2 + 3x, (1, -1)\big) + \big(x, (0, 5)\big) = \big(2 + 4x, (1, 4)\big)$.

Proposition (Prop. 35, 36 / LADR 3.76, HW4 P5)

Let $V_{(i)} = \{(0, \ldots, 0, v_i, 0, \ldots, 0) : v_i \in V_i\}$ (entry v_i in position i, zeros elsewhere); $V_{(i)}$ is a copy of V_i. Then

$$V_1 \times \cdots \times V_m = V_{(1)} \oplus \cdots \oplus V_{(m)},$$

and if each V_i is finite-dimensional, $\dim(V_1 \times \cdots \times V_m) = \dim V_1 + \cdots + \dim V_m$.

Proof. $(v_1, \ldots, v_m) = (v_1, 0, \ldots, 0) + \cdots + (0, \ldots, 0, v_m)$, and this is the only way to write it as a sum of one vector from each $V_{(i)}$, because the entries determine the pieces. For the dimension, take two spaces: if $v_1, \ldots, v_k$ is a basis of V and $w_1, \ldots, w_l$ a basis of W, then

$$(v_1, 0), \ldots, (v_k, 0),\ (0, w_1), \ldots, (0, w_l)$$

is a basis of $V \times W$. It spans, since $(v, w) = (v, 0) + (0, w)$ and each part is a combination of the listed vectors. It is independent, since $\sum a_i(v_i, 0) + \sum b_j(0, w_j) = \big(\sum a_i v_i, \sum b_j w_j\big) = (0, 0)$ forces all $a_i = 0$ and all $b_j = 0$. So $\dim(V \times W) = k + l$; repeat for more factors. □

Because of this, the product is also called the **external direct sum**. The dimension of a product is the *sum* of the dimensions (compare $\dim \mathcal{L}(V, W) = (\dim V)(\dim W)$).

Proposition (Prop. 37 / LADR 3.93)

For subspaces $U_1, \ldots, U_m$ of V, the map

$$\Gamma\colon U_1 \times \cdots \times U_m \to U_1 + \cdots + U_m, \qquad \Gamma(u_1, \ldots, u_m) = u_1 + \cdots + u_m$$

is linear and surjective, and it is injective iff the sum $U_1 + \cdots + U_m$ is direct. Consequently $\dim(U_1 + \cdots + U_m) \leq \dim U_1 + \cdots + \dim U_m$, with equality iff the sum is direct.

Proof. Surjective: every element of $U_1 + \cdots + U_m$ is some $u_1 + \cdots + u_m$. Injective means $u_1 + \cdots + u_m = 0$ only when every $u_i = 0$, which is the test for a direct sum (Section 1.1). For the dimensions, rank–nullity gives $\dim(U_1 + \cdots + U_m) = \dim(U_1 \times \cdots \times U_m) - \dim \operatorname{null} \Gamma$, and $\dim(U_1 \times \cdots \times U_m) = \sum \dim U_i$. □

For two subspaces, $\operatorname{null}\Gamma = \{(u,-u) : u \in U \cap W\}$, a copy of $U \cap W$. Rank–nullity then gives the formula $\dim(U+W) = \dim U + \dim W - \dim(U \cap W)$.

Example

In $\mathbb{R}^3$ let $U = \operatorname{span}(e_1, e_2)$ and $W = \operatorname{span}(e_2, e_3)$. $\Gamma\colon U \times W \to \mathbb{R}^3$ is surjective, and $\operatorname{null}\Gamma = \{(te_2, -te_2) : t \in \mathbb{R}\}$ has dimension 1. So $\dim(U+W) = 4 - 1 = 3$, and the sum is not direct.

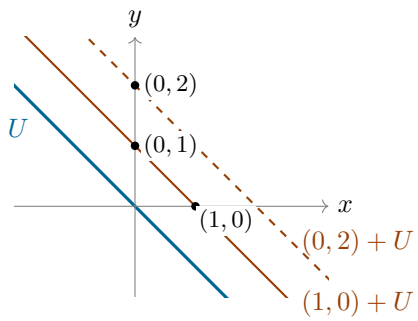
Traps.

- $V \times W$ is a new space of pairs. It is not $V \cup W$, and it is not $V + W$ (which only makes sense when V and W sit inside one larger space).
- $\dim(V \times W) = \dim V + \dim W$, not $(\dim V)(\dim W)$.

§1.3 Affine subsets and quotient spaces

The idea. The solutions of $Tx = b$ with $b \neq 0$ do not form a subspace (they do not contain 0), but they form a subspace shifted away from 0. Such shifted subspaces are called affine subsets. If we fix one subspace U and collect all of its shifts, these shifts can themselves be added and scaled: that is the quotient space V/U. In V/U we stop distinguishing two vectors whose difference lies in U.

Definition (Def. 22, 23 / LADR 3.95, 3.97)**.** For a subspace $U \subseteq V$ and $v \in V$, the **translate** $v + U = \{v + u : u \in U\}$ is an **affine subset** of V, and it is **parallel** to U. Two affine subsets are parallel if they are parallel to the same subspace.



$U = \operatorname{span}\big((-1,1)\big)$ is the line $x + y = 0$. The line $x + y = 1$ is the affine subset $(1,0) + U$ (Ex. 41); it is also $(0,1) + U$, because $(1,0) - (0,1) = (1,-1) \in U$. The dashed line $x + y = 2$ is $(0,2) + U$. Each of these lines is one element of the quotient space $\mathbb{R}^2/U$.

Example (Ex. 41, 42; solution sets)

- The line $\{(x,y) : x + y = 1\}$ in $\mathbb{R}^2$ is $(1,0) + \operatorname{span}\big((-1,1)\big)$.
- Indefinite integrals: $\int x^2\,dx = \frac{1}{3}x^3 + C$ is the affine subset $\frac{1}{3}x^3 + \mathbb{R}$ of $\mathcal{P}(\mathbb{R})$, parallel to the subspace $\mathbb{R}$ of constant polynomials.
- If $Tx_0 = b$, then $\{x : Tx = b\} = x_0 + \operatorname{null} T$: indeed $Tx = b$ iff $T(x - x_0) = 0$ iff $x - x_0 \in \operatorname{null} T$.

Proposition (Prop. 38)

An affine subset is parallel to only one subspace: if $v + U = x + W$, then $U = W$.

Proof. Since $v \in x + W$, write $v = x + w_0$ with $w_0 \in W$. For $u \in U$, $v + u \in x + W$, so $v + u = x + w$ for some $w \in W$, and $u = w - w_0 \in W$. So $U \subseteq W$; the same argument with the roles switched gives $W \subseteq U$. □

Proposition (Prop. 39 / LADR 3.101, HW4 P6)

For a subspace U and $v, w \in V$, these are equivalent:

1. $v - w \in U$;
2. $v + U = w + U$;
3. $(v + U) \cap (w + U) \neq \varnothing$.

Also, $v + U$ is a subspace iff $v \in U$, and then $v + U = U$.

Proof. (1)⇒(2): for $u \in U$, $v + u = w + \big((v - w) + u\big) \in w + U$, so $v + U \subseteq w + U$; the same argument gives the other inclusion. (2)⇒(3): $v = v + 0$ lies in both. (3)⇒(1): if $v + u_1 = w + u_2$, then $v - w = u_2 - u_1 \in U$. For the last statement: if $v + U$ is a subspace it contains 0, so $0 = v + u$ for some $u \in U$ and $v = -u \in U$. If $v \in U$, then $v - 0 \in U$, so $v + U = 0 + U = U$ by (1)⇒(2). □

So two translates are either equal or disjoint, like the parallel lines in the figure.

Definition (Def. 24)**.** The **quotient space** V/U is the set of all affine subsets of V parallel to U, with

$$(v + U) + (w + U) = (v + w) + U, \qquad \lambda(v + U) = (\lambda v) + U.$$

Its zero is $0 + U = U$.

Why these operations make sense ("well-defined"). The same translate has many names: $v + U = v' + U$ whenever $v - v' \in U$. The formulas above use a name v, so we must check that the answer does not depend on which name we use. If $v_1 + U = v_2 + U$ and $w_1 + U = w_2 + U$, then $v_1 - v_2 \in U$ and $w_1 - w_2 \in U$, so $(v_1 + w_1) - (v_2 + w_2) \in U$ and the two sums are the same translate. Similarly $\lambda v_1 - \lambda v_2 = \lambda(v_1 - v_2) \in U$. This kind of check is needed every time a map out of V/U is defined by a formula in v.

Example

In $\mathbb{R}^2$ with $U = \operatorname{span}(e_2)$ (the y-axis), each element of $\mathbb{R}^2/U$ is a vertical line $\{x = c\}$. $(1, 2) + U = (1, 4) + U$ because $(0, -2) \in U$: in $\mathbb{R}^2/U$ only the x-coordinate is remembered.

Proposition (Basis of a quotient)

Let $u_1, \ldots, u_m$ be a basis of U, extended to a basis $u_1, \ldots, u_m, v_1, \ldots, v_k$ of V. Then $v_1 + U, \ldots, v_k + U$ is a basis of V/U. In particular $\dim V/U = \dim V - \dim U$.

Proof. Spanning: write $v = \sum a_i u_i + \sum b_j v_j$; then $v + U = \sum b_j (v_j + U)$, since $\sum a_i u_i \in U$. Independence: if $\sum b_j (v_j + U) = 0 + U$, then $\sum b_j v_j \in U$, so $\sum b_j v_j = \sum c_i u_i$ for some c_i. The whole list $u_1, \ldots, u_m, v_1, \ldots, v_k$ is independent, so all $b_j = 0$. □

Traps.

- An element of V/U is a set $v + U$, not a vector of V. V/U is not a subspace of V.
- $v + U = w + U$ does not mean $v = w$; it means $v - w \in U$.
- Every formula on V/U written in terms of v needs a well-definedness check.

§1.4 Quotient maps and the isomorphism theorems

The idea. Every surjective linear map $T\colon V \to W$ makes W look like a quotient of V: two vectors have the same image exactly when their difference is in $\operatorname{null} T$, so W "is" $V/\operatorname{null} T$. The isomorphism theorems make this precise, and they are the standard way to identify a quotient space with a familiar space.

Definition (Def. 25 / LADR 3.104)**.** The **quotient map** is $\pi\colon V \to V/U$, $\pi(v) = v + U$. It is linear and surjective, and $\operatorname{null} \pi = U$, since $v + U = 0 + U$ iff $v \in U$.

Proposition (Prop. 40 / LADR 3.105)
If V is finite-dimensional, $\dim V/U = \dim V - \dim U$.

Proof. Rank–nullity for π: $\dim V = \dim \operatorname{null} \pi + \dim \operatorname{range} \pi = \dim U + \dim V/U$. □

Theorem (Prop. 41 / LADR 3.107: first isomorphism theorem)
For $T \in \mathcal{L}(V, W)$, the map

$$\bar{T}\colon V/\operatorname{null} T \to \operatorname{range} T, \qquad \bar{T}(v + \operatorname{null} T) = Tv$$

is a well-defined isomorphism, and $T = \iota \circ \bar{T} \circ \pi$, where $\pi\colon V \to V/\operatorname{null} T$ is the quotient map and $\iota\colon \operatorname{range} T \to W$ is the inclusion $\iota(w) = w$.

$$\begin{array}{ccc} V & \xrightarrow{\ T\ } & W \\ {\scriptstyle\pi}\downarrow & & \uparrow{\scriptstyle\iota} \\ V/\operatorname{null} T & \xrightarrow{\ \bar{T}\ } & \operatorname{range} T \end{array}$$

Proof. Well defined: if $v + \operatorname{null} T = v' + \operatorname{null} T$, then $v - v' \in \operatorname{null} T$, so $Tv = Tv'$. Linear: from linearity of T. Injective: $\bar{T}(v + \operatorname{null} T) = 0$ means $Tv = 0$, so $v \in \operatorname{null} T$ and $v + \operatorname{null} T$ is the zero of $V/\operatorname{null} T$. Surjective: every element of $\operatorname{range} T$ is some $Tv = \bar{T}(v + \operatorname{null} T)$. □

How to use it. To show $V/U \cong W$, find a linear map $T\colon V \to W$ that is surjective and has $\operatorname{null} T = U$ exactly. Then $v + U \mapsto Tv$ is the isomorphism.

Example (Ex. 43)
Differentiation $D\colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ is surjective with $\operatorname{null} D = \mathbb{R}$ (the constants). So $\mathcal{P}(\mathbb{R})/\mathbb{R} \cong \mathcal{P}(\mathbb{R})$ by $p + \mathbb{R} \mapsto p'$; the inverse sends p to its indefinite integral $\int p\,dx$, an element of $\mathcal{P}(\mathbb{R})/\mathbb{R}$.

Example (A quotient of $\mathcal{P}_3(\mathbb{R})$)
Let $T\colon \mathcal{P}_3(\mathbb{R}) \to \mathbb{R}^2$, $T(p) = \big(p(1), p'(1)\big)$. It is surjective: $T(1) = (1, 0)$ and $T(x - 1) = (0, 1)$. So $\dim \operatorname{null} T = 4 - 2 = 2$, and $\operatorname{null} T = \{p : p(1) = p'(1) = 0\} = \operatorname{span}\big((x-1)^2,\ x(x-1)^2\big)$ (both are in it, and they are independent). By the first isomorphism theorem, $\mathcal{P}_3(\mathbb{R})/\operatorname{null} T \cong \mathbb{R}^2$ via $p + \operatorname{null} T \mapsto \big(p(1), p'(1)\big)$. A basis of the quotient is $1 + \operatorname{null} T$, $(x - 1) + \operatorname{null} T$, the preimages of $(1, 0)$ and $(0, 1)$.

Proposition (Prop. 42; HW5 P2: induced maps)
Let $T \in \mathcal{L}(V, W)$, and let $U_1 \subseteq V$, $U_2 \subseteq W$ be subspaces with $T(U_1) \subseteq U_2$. Then

$$\bar{T}\colon V/U_1 \to W/U_2, \qquad \bar{T}(v + U_1) = Tv + U_2$$

is a well-defined linear map. In particular (take $U_2 = \{0\}$): if $U \subseteq \operatorname{null} T$, there is exactly one linear $\tilde{T}\colon V/U \to W$ with $\tilde{T}(v + U) = Tv$, that is $T = \tilde{T} \circ \pi$, and $\operatorname{null} \tilde{T} = \{v + U : v \in \operatorname{null} T\} = (\operatorname{null} T)/U$.

Proof. Well defined: if $v + U_1 = v' + U_1$, then $v - v' \in U_1$, so $Tv - Tv' = T(v - v') \in U_2$ and $Tv + U_2 = Tv' + U_2$. Linearity comes from T. For $\tilde{T}$: it is unique because its value on every $v + U$ is forced to be Tv. And $\tilde{T}(v + U) = 0$ iff $Tv = 0$ iff $v \in \operatorname{null} T$. □

The quotient operator T/U of Section 1.10 is the case $V = W$ and $U_1 = U_2 = U$.

Theorem (Prop. 43: second isomorphism theorem)
For subspaces U, W of V: $U/(U \cap W) \cong (U + W)/W$, via $u + (U \cap W) \mapsto u + W$. In finite dimensions this gives $\dim U - \dim(U \cap W) = \dim(U + W) - \dim W$.

Proof. The map $S\colon U \to (U+W)/W$, $S(u) = u+W$, is linear. It is surjective: an element $u+w+W$ of $(U+W)/W$ equals $u+W$, since $(u+w)-u \in W$. Its null space: $u+W = 0+W$ iff $u \in W$, so $\operatorname{null} S = U \cap W$. Apply the first isomorphism theorem to S. □

Theorem (Prop. 44: third isomorphism theorem)
For subspaces $W \subseteq U \subseteq V$: $(V/W)/(U/W) \cong V/U$, via $(v+W)+U/W \mapsto v+U$. (Here $U/W = \{u+W : u \in U\}$, a subspace of V/W.)

Proof. The identity map of V sends W into U, so by Prop. 42 it induces $S\colon V/W \to V/U$, $S(v+W) = v+U$. S is surjective, and $S(v+W) = 0+U$ iff $v \in U$ iff $v+W \in U/W$. So $\operatorname{null} S = U/W$; apply the first isomorphism theorem. □

These are used in proofs by induction on dimension: if $W \neq \{0\}$, then V/W is smaller than V, and the third isomorphism theorem relates quotients of V/W to quotients of V.

Traps.

- The first isomorphism theorem gives $V/\operatorname{null} T \cong \operatorname{range} T$, which is W only when T is surjective.
- To prove $V/U \cong W$ with it, you need $\operatorname{null} T = U$ exactly, not just $U \subseteq \operatorname{null} T$.
- A map out of a quotient defined by "$v+U \mapsto$ (formula in v)" is only a map after the well-definedness check.

§1.5 Linear functionals and the dual space

The idea. Often we want one number from a vector: its second coordinate, the value of a polynomial at 2, the integral of a function from 0 to 1. When that number depends linearly on the vector, the rule is called a linear functional. The dual space is the set of all such rules. Studying it gives a second view of subspaces and linear maps, and it explains why transposes appear.

Notation on the exam. The notes write V' and T' for the dual space and the dual map. Hammond's practice midterm writes V^* and T^* for the same things; they mean exactly the same.

Definition (Def. 26)**.** A **linear functional** on V is a linear map $\varphi\colon V \to \mathbb{F}$. The **dual space** of V is

$$V' = \mathcal{L}(V, \mathbb{F}) = \{\text{all linear functionals on } V\}.$$

It is a vector space with $(\varphi+\psi)(v) = \varphi(v)+\psi(v)$ and $(c\varphi)(v) = c\,\varphi(v)$.

The prime in V' has nothing to do with derivatives. Functionals are usually written with Greek letters: φ (phi), ψ (psi).

Example

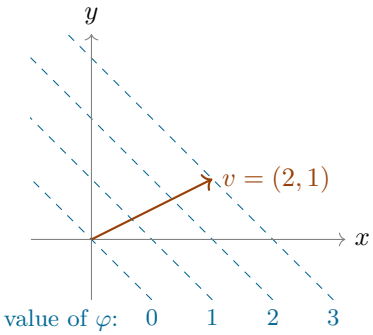
- On $\mathbb{R}^3$: $\varphi(x,y,z) = 2x - y + 3z$.
- On $\mathcal{P}_2(\mathbb{R})$: $p \mapsto p(2)$, $p \mapsto p'(0)$, $p \mapsto \int_0^1 p(x)\,dx$. Each is linear in p.
- Not functionals: $(x,y) \mapsto x+1$ (it sends 0 to 1, and a linear map sends 0 to 0); $p \mapsto p(2)^2$ (doubling p multiplies the output by 4).

Functionals on $\mathbb{F}^n$ are row vectors (Ex. 44). If $\varphi \in (\mathbb{F}^n)'$ and $a_j = \varphi(e_j)$, then for $x = (x_1, \dots, x_n)$, linearity gives

$$\varphi(x) = \varphi(x_1 e_1 + \cdots + x_n e_n) = a_1 x_1 + \cdots + a_n x_n = \begin{pmatrix} a_1 & \cdots & a_n \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

So a functional on $\mathbb{F}^n$ is the same thing as a row vector, and applying it is multiplying the row by the column. The dual basis of $e_1, \dots, e_n$ (defined below) is the list of rows $(1\ 0\ \cdots\ 0)$, $\dots$, $(0\ \cdots\ 0\ 1)$.

A picture of a functional. In $\mathbb{R}^2$, the vectors where φ takes a fixed value c form a line, and different values give parallel, evenly spaced lines. $\varphi(v)$ tells you which of these lines v ends on.



$\varphi(x, y) = x + y$. Each dashed line is where φ takes one value. $v = (2, 1)$ ends on the line $\varphi = 3$, so $\varphi(v) = 3$. The line $\varphi = 0$ is $\operatorname{null} \varphi$.

Proposition (HW5 P3)

Every functional $\varphi \in V'$ is either the zero functional or surjective onto $\mathbb{F}$. So if $\varphi \neq 0$, then $\dim \operatorname{null} \varphi = \dim V - 1$.

Proof. $\operatorname{range} \varphi$ is a subspace of $\mathbb{F}$, and $\dim \mathbb{F} = 1$, so the range is $\{0\}$ or all of $\mathbb{F}$. If $\varphi \neq 0$, the range is $\mathbb{F}$, which has dimension 1, so rank–nullity gives $\dim \operatorname{null} \varphi = \dim V - 1$. □

§1.6 The dual basis

The idea. A basis $v_1, \ldots, v_n$ lets you write each vector as $v = a_1 v_1 + \cdots + a_n v_n$. The rule "$v \mapsto a_j$" (read off the jth coefficient) is a linear functional. These n functionals form the dual basis.

Definition (Def. 27)**.** Let $v_1, \ldots, v_n$ be a basis of V. The **dual basis** is the list $\varphi_1, \ldots, \varphi_n \in V'$ defined by

$$\varphi_j(v_k) = \begin{cases} 1 & \text{if } j = k, \\ 0 & \text{if } j \neq k. \end{cases}$$

(A linear map is determined by its values on a basis, so these conditions define each φ_j.)

Why this is "the jth coefficient": if $v = a_1 v_1 + \cdots + a_n v_n$, then $\varphi_j(v) = a_1 \varphi_j(v_1) + \cdots + a_n \varphi_j(v_n) = a_j$, because only the $k = j$ term survives.

Proposition (Prop. 45 / LADR 3.116, HW5 P4)

The dual basis is a basis of V', so $\dim V' = \dim V$. For every $v \in V$ and every $\psi \in V'$:

$$v = \sum_{j=1}^{n} \varphi_j(v)\, v_j, \qquad \psi = \sum_{j=1}^{n} \psi(v_j)\, \varphi_j.$$

Proof. $\dim V' = \dim \mathcal{L}(V, \mathbb{F}) = (\dim V)(\dim \mathbb{F}) = n$, so it is enough to show $\varphi_1, \ldots, \varphi_n$ is linearly independent. Suppose $c_1 \varphi_1 + \cdots + c_n \varphi_n = 0$ (the zero functional). Apply both sides to v_k: the left side is c_k and the right side is 0. So every $c_k = 0$. The first formula is the "jth coefficient" fact above. For the second, both sides are linear functionals, and at each basis vector v_k both give $\psi(v_k)$. Two linear maps that agree on a basis are equal. □

The second formula is how you write a given functional in the dual basis: *evaluate it on the original basis vectors.*

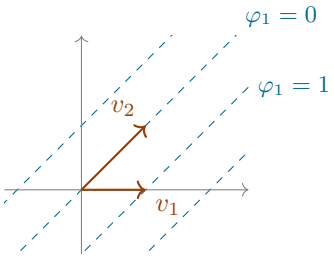
How to compute a dual basis in $\mathbb{F}^n$. Put $v_1, \ldots, v_n$ as the columns of a matrix B. Then the dual basis is the list of rows of B^{-1}. Reason: the entry in row j, column k of $B^{-1}B$ is (row j of B^{-1}) times (column k of B), which is $\varphi_j(v_k)$. And $B^{-1}B = I$ says exactly that this is 1 when $j = k$ and 0 otherwise.

Example (A dual basis in $\mathbb{R}^2$)

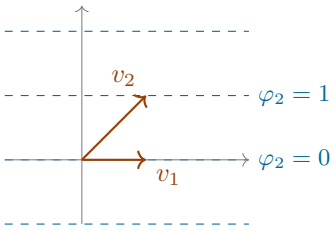
Take $v_1 = (1,0)$ and $v_2 = (1,1)$. Then

$$B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad B^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix},$$

so $\varphi_1(x,y) = x - y$ and $\varphi_2(x,y) = y$. Check: $\varphi_1(v_1) = 1$, $\varphi_1(v_2) = 0$, $\varphi_2(v_1) = 0$, $\varphi_2(v_2) = 1$.



level lines of $\varphi_1 = x - y$



level lines of $\varphi_2 = y$

φ_1 is 0 on the line through v_2 and 1 on the parallel line through v_1. φ_2 is 0 on the line through v_1 and 1 on the parallel line through v_2. The direction of the level lines of φ_1 is set by v_2: changing v_2 changes φ_1, even though v_1 stays the same.

Example (A dual basis in $\mathbb{R}^3$)

For $v_1 = (1,0,0)$, $v_2 = (1,1,0)$, $v_3 = (1,1,1)$:

$$B = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad B^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix},$$

so the dual basis is $\varphi_1 = x - y$, $\varphi_2 = y - z$, $\varphi_3 = z$. To write $\psi(x,y,z) = x + y + z$ in this basis, evaluate ψ on v_1, v_2, v_3: the values are $1, 2, 3$, so $\psi = \varphi_1 + 2\varphi_2 + 3\varphi_3$. Check: $(x - y) + 2(y - z) + 3z = x + y + z$.

Example (A dual basis of polynomials)

For the basis $1, x, x^2$ of $\mathcal{P}_2(\mathbb{R})$, the dual basis is

$$\varphi_1(p) = p(0), \qquad \varphi_2(p) = p'(0), \qquad \varphi_3(p) = \tfrac{1}{2}p''(0).$$

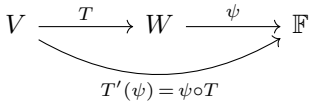
Check: for $p = a + bx + cx^2$, these give a, b, c, the three coefficients.

Traps.

- φ_j depends on the *whole* basis, not only on v_j. In the $\mathbb{R}^3$ example, $\varphi_1 = x - y$, not "the x-coordinate".
- V and V' have the same dimension, so they are isomorphic. But there is no preferred isomorphism: $v_j \mapsto \varphi_j$ depends on which basis you chose.

§1.7 The dual map

The idea. A linear map $T\colon V \to W$ turns each functional on W into a functional on V: to evaluate it at v, first move v to Tv, then apply the functional on W.



Definition (Def. 28). For $T \in \mathcal{L}(V,W)$, the **dual map** is $T'\colon W' \to V'$, $T'(\psi) = \psi \circ T$. That is, $\big(T'(\psi)\big)(v) = \psi(Tv)$.

The direction reverses: T goes from V to W, and T' goes from W' to V'. Two linearity facts: $T'(\psi) = \psi \circ T$ is a linear functional on V (a composition of linear maps), and T' itself is linear, since $T'(\psi_1 + c\psi_2) = (\psi_1 + c\psi_2) \circ T = T'(\psi_1) + cT'(\psi_2)$ (evaluate both sides at any v).

Example (Ex. 45)
$T\colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$, $Tp = xp$, and $\varphi(p) = p(2)$. Then $\big(T'(\varphi)\big)(p) = \varphi(xp) = 2p(2)$. So $T'(\varphi) = 2\varphi$.

Example (Computing T' in a basis)
Let $T(x, y) = (x + 2y,\ 3x + 4y)$ on $\mathbb{R}^2$, so $\mathcal{M}(T) = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right)$, and let $\varphi_1(x, y) = x$, $\varphi_2(x, y) = y$ (the dual basis of e_1, e_2). Then

$$T'(\varphi_1)(x, y) = \varphi_1\big(T(x, y)\big) = x + 2y, \quad \text{so } T'(\varphi_1) = \varphi_1 + 2\varphi_2;$$

$$T'(\varphi_2)(x, y) = 3x + 4y, \quad \text{so } T'(\varphi_2) = 3\varphi_1 + 4\varphi_2.$$

The columns of $\mathcal{M}(T')$ are $(1, 2)$ and $(3, 4)$: $\mathcal{M}(T') = \left(\begin{smallmatrix} 1 & 3 \\ 2 & 4 \end{smallmatrix}\right)$, the transpose of $\mathcal{M}(T)$.

Example (Ex. 46)
For $A = \left(\begin{smallmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{smallmatrix}\right)\colon \mathbb{R}^3 \to \mathbb{R}^2$, write functionals as rows. Then A' sends the row $(x \ \ y)$ to the row $(x \ \ y)A$: first multiply by A, then by the row.

Proposition (Prop. 47 / LADR 3.118, 3.120)
The map $T \mapsto T'$ is linear: $(S + T)' = S' + T'$ and $(cT)' = cT'$. It reverses products: $(ST)' = T'S'$.

Proof. Linearity: $\big((S + T)'\psi\big)(v) = \psi(Sv + Tv) = \psi(Sv) + \psi(Tv)$; scalars are the same. Products: $(ST)'\psi = \psi \circ S \circ T = (S'\psi) \circ T = T'(S'\psi)$. □

Proposition (Prop. 49 / LADR 3.132)
Choose bases $v_1, \dots, v_n$ of V and $w_1, \dots, w_m$ of W, and use their dual bases for V' and W'. Then $\mathcal{M}(T') = \mathcal{M}(T)^T$, the transpose ($(A^T)_{ij} = A_{ji}$: rows become columns).

Proof. Let $\mathcal{M}(T) = (a_{ij})$, so $Tv_k = \sum_l a_{lk} w_l$. Write $\psi_1, \dots, \psi_m$ for the dual basis of $w_1, \dots, w_m$ and $\varphi_1, \dots, \varphi_n$ for the dual basis of $v_1, \dots, v_n$. Then $\big(T'(\psi_j)\big)(v_k) = \psi_j(Tv_k) = a_{jk}$. By the formula $\psi = \sum_k \psi(v_k)\varphi_k$, $T'(\psi_j) = \sum_k a_{jk}\varphi_k$. So column j of $\mathcal{M}(T')$ is $(a_{j1}, \dots, a_{jn})$, which is row j of $\mathcal{M}(T)$. □

Together with $(ST)' = T'S'$ this gives $(AB)^T = B^T A^T$ for matrices.

Traps.

- T' goes from W' to V', not from V' to W'.
- The matrix of T' is the transpose only when you use the *dual* bases of the bases you used for T.
- To compute $T'(\psi)$, apply T first and ψ second: $T'(\psi)(v) = \psi(Tv)$.

§1.8 The double dual

The idea. $V'' = (V')'$ is the set of linear functionals on V'. Each vector v gives one without any choices: "take a functional φ and evaluate it at v".

Proposition (Prop. 46, 48 / LADR 3F)
Define $j_V\colon V \to V''$ by $j_V(v)(\varphi) = \varphi(v)$. Then j_V is linear and injective, and if V is finite-dimensional it is an isomorphism. For every $T \in \mathcal{L}(V, W)$, $T'' \circ j_V = j_W \circ T$, where $T'' = (T')'\colon V'' \to W''$.

Proof. Injective: suppose $v \neq 0$. Extend v to a basis $v, v_2, \dots$ of V, and let φ be the functional with $\varphi(v) = 1$ and $\varphi = 0$ on the other basis vectors. Then $j_V(v)(\varphi) = 1 \neq 0$, so $j_V(v) \neq 0$. If V is finite-dimensional, $\dim V'' = \dim V' = \dim V$,

and an injective linear map between spaces of the same finite dimension is invertible. Compatibility: for $\psi \in W'$, $\big(T''(j_V v)\big)(\psi) = j_V(v)(T'\psi) = (T'\psi)(v) = \psi(Tv) = j_W(Tv)(\psi)$. □

The notes call j_V **canonical**: it is defined by the same formula for every space, without choosing a basis, and it satisfies $T'' \circ j_V = j_W \circ T$ for every linear map T. Contrast this with $V \cong V'$, which needs a basis.

Consequence used often: if $\varphi(v) = 0$ for every $\varphi \in V'$, then $v = 0$. Equivalently, if $\varphi(v) = \varphi(w)$ for every φ, then $v = w$.

§1.9 Annihilators

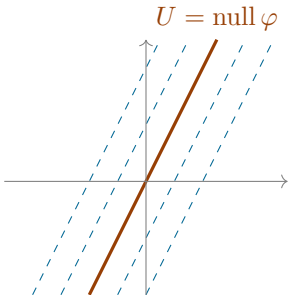
Throughout this subsection, V and W are finite-dimensional.

The idea. Given a subspace U of V, ask which functionals are 0 on every vector of U. They form a subspace of V', the annihilator. Many statements about subspaces of V become statements about subspaces of V' this way.

Definition (Def. 29)**.** For a subspace $U \subseteq V$, the **annihilator** of U is

$$U^0 = \{\varphi \in V' : \varphi(u) = 0 \text{ for every } u \in U\}.$$

U^0 is a subspace of V': if φ and ψ are 0 on U, so are $\varphi + \psi$ and $c\varphi$.



$U = \operatorname{span}\big((1,2)\big)$ in $\mathbb{R}^2$. A functional is 0 on U exactly when its zero line is U (or it is the zero functional). The functional $\varphi(x,y) = 2x - y$ has zero line U; its other level lines (dashed) are parallel to U. So $U^0 = \operatorname{span}(\varphi)$, and $\dim U^0 = 2 - 1 = 1$.

Example (Ex. 47, HW5 P6)

$U = \operatorname{span}\big((1,1,0),(0,1,1)\big) \subseteq \mathbb{R}^3$. Write a functional as a row (a,b,c), meaning $\varphi(x,y,z) = ax + by + cz$. It is 0 on U iff it is 0 on both spanning vectors:

$$a + b = 0, \qquad b + c = 0.$$

So $b = -a$, $c = a$, and $U^0 = \operatorname{span}\big((1,-1,1)\big)$, the functional $x - y + z$.

Recipe in $\mathbb{F}^n$. Write a basis of U as the rows of a matrix M. Then U^0 is the null space of M, with each solution read as a row vector.

Example

In $\mathcal{P}_2(\mathbb{R})$ let $U = \operatorname{span}(1, x) = \mathcal{P}_1(\mathbb{R})$. A functional ψ is 0 on U iff $\psi(1) = \psi(x) = 0$. In the dual basis $\varphi_1, \varphi_2, \varphi_3$ of $1, x, x^2$, $\psi = \psi(1)\varphi_1 + \psi(x)\varphi_2 + \psi(x^2)\varphi_3$, so $U^0 = \operatorname{span}(\varphi_3)$: the functional "coefficient of x^2".

Proposition (LADR 3.125)

$\dim U^0 = \dim V - \dim U$.

Proof. Extend a basis $u_1, \ldots, u_m$ of U to a basis $u_1, \ldots, u_m, v_{m+1}, \ldots, v_n$ of V, and let $\varphi_1, \ldots, \varphi_n$ be the dual basis. Any $\psi \in V'$ is $\psi = \sum_j c_j \varphi_j$ with $c_j = \psi(j\text{th basis vector})$. It is 0 on U iff it is 0 on $u_1, \ldots, u_m$, that is, iff $c_1 = \cdots = c_m = 0$. So $U^0 = \operatorname{span}(\varphi_{m+1}, \ldots, \varphi_n)$, which has dimension $n - m$. □

Each of the $\dim U$ independent vectors of U imposes one condition on ψ, and each condition lowers the dimension by one.

Extending functionals. Every functional on a subspace U extends to a functional on V: extend a basis of U to a basis of V and let the functional be 0 on the added vectors. So the restriction map $V' \to U'$, $\varphi \mapsto \varphi|_U$, is surjective. Its null space is U^0, so rank–nullity gives $\dim V' = \dim U^0 + \dim U'$, which is again $\dim U^0 = \dim V - \dim U$.

Proposition (Prop. 50 / LADR 3.128(a), 3.130(b))

For $T \in \mathcal{L}(V, W)$:

1. $\operatorname{null} T' = (\operatorname{range} T)^0$;
2. $\operatorname{range} T' = (\operatorname{null} T)^0$.

Proof. (1) $T'\psi = 0$ means $\psi(Tv) = 0$ for every $v \in V$, which says exactly that ψ is 0 on every vector of $\operatorname{range} T$. (2) "$\subseteq$": if $\varphi = T'\psi$ and $v \in \operatorname{null} T$, then $\varphi(v) = \psi(Tv) = \psi(0) = 0$. "$\supseteq$": let φ be 0 on $\operatorname{null} T$. We need $\psi \in W'$ with $\psi \circ T = \varphi$. Define ψ on $\operatorname{range} T$ by $\psi(Tv) = \varphi(v)$. This does not depend on which v we use: if $Tv = Tv'$, then $v - v' \in \operatorname{null} T$, so $\varphi(v) = \varphi(v')$. Extend a basis of $\operatorname{range} T$ to a basis of W and let ψ be 0 on the added vectors. Then $\psi(Tv) = \varphi(v)$ for all v, so $T'\psi = \varphi$. □

Corollary (Prop. 51 / LADR 3.129–3.133)

T is injective iff T' is surjective, and T is surjective iff T' is injective. Also $\operatorname{rank} T' = \operatorname{rank} T$. For matrices: $\operatorname{rank} A^T = \operatorname{rank} A$, so the number of independent rows equals the number of independent columns.

Proof. T surjective $\iff \operatorname{range} T = W \iff (\operatorname{range} T)^0 = \{0\} \iff \operatorname{null} T' = \{0\}$. T injective $\iff \operatorname{null} T = \{0\} \iff (\operatorname{null} T)^0 = V' \iff \operatorname{range} T' = V'$. Rank: $\dim \operatorname{range} T' = \dim(\operatorname{null} T)^0 = \dim V - \dim \operatorname{null} T = \dim \operatorname{range} T$. □

Corollary (LADR 3.128(b))

For $T \in \mathcal{L}(V, W)$: $\dim \operatorname{null} T' = \dim \operatorname{null} T + \dim W - \dim V$.

Proof. $\dim \operatorname{null} T' = \dim(\operatorname{range} T)^0 = \dim W - \dim \operatorname{range} T = \dim W - (\dim V - \dim \operatorname{null} T)$. □

Proposition (Separating functional)

If $W \subseteq V$ is a subspace and $v \notin W$, there is $\varphi \in V'$ with $\varphi = 0$ on W and $\varphi(v) = 1$. In particular, if $W \neq V$ then $W^0 \neq \{0\}$.

Proof. Since $v \notin W$, a basis of W followed by v is linearly independent. Extend it to a basis of V, and let φ be 1 on v and 0 on every other basis vector. □

Proposition (Rules for annihilators)

For subspaces U, W of V:

1. $U^0 = V'$ iff $U = \{0\}$, and $U^0 = \{0\}$ iff $U = V$.
2. $U \subseteq W$ iff $W^0 \subseteq U^0$ (bigger subspace, smaller annihilator).
3. $(U + W)^0 = U^0 \cap W^0$ and $(U \cap W)^0 = U^0 + W^0$.
4. $(V/U)' \cong U^0$.
5. $U = \{v \in V : \varphi(v) = 0 \text{ for every } \varphi \in U^0\}$: a subspace is determined by its annihilator.

Proof. (1) From $\dim U^0 = \dim V - \dim U$. (2) If $U \subseteq W$, a functional that is 0 on W is 0 on U. Conversely, if $u \in U$ but $u \notin W$, the separating functional is in W^0 but not in U^0. (3) A functional is 0 on every $u + w$ iff it is 0 on U and

on W. For the second identity, $U^0 + W^0 \subseteq (U \cap W)^0$ is clear (a sum of a functional that is 0 on U and one that is 0 on W is 0 on $U \cap W$), and with $n = \dim V$ the dimensions agree:

$$\begin{aligned}\dim(U^0 + W^0) &= \dim U^0 + \dim W^0 - \dim(U^0 \cap W^0) \\ &= (n - \dim U) + (n - \dim W) - \big(n - \dim(U + W)\big) = n - \dim(U \cap W).\end{aligned}$$

(4) For $\varphi \in U^0$ put $\tilde{\varphi}(v + U) = \varphi(v)$. This does not depend on the choice of v, because φ is 0 on U. The map $\varphi \mapsto \tilde{\varphi}$ from U^0 to $(V/U)'$ is linear and injective, and both spaces have dimension $\dim V - \dim U$, so it is an isomorphism.
(5) Every $u \in U$ satisfies $\varphi(u) = 0$ for $\varphi \in U^0$. If $v \notin U$, the separating functional is in U^0 and is 1 at v. □

Traps.

- U^0 is a set of functionals, a subspace of V', not a subspace of V. Its basis is a list of functionals, not vectors.
- U^0 is not the "orthogonal complement". That notion needs an inner product, which comes after this midterm.
- $(V/U)' \cong U^0$, not U'.

§1.10 Invariant subspaces

The idea. To understand an operator $T \in \mathcal{L}(V)$, look for pieces of V that T maps into themselves. Then T can be studied one piece at a time, and each piece is smaller than V. Almost every proof from here to the end of the midterm material works this way.

Definition (Def. 30 / LADR 5.2)**.** A subspace $U \subseteq V$ is **invariant** under T if $T(U) \subseteq U$, that is, $Tu \in U$ for every $u \in U$.

When U is invariant, two new operators are defined:

- the **restriction** $T|_U \in \mathcal{L}(U)$, which is T used only on vectors of U (it needs $Tu \in U$ to be an operator on U);
- the **quotient operator** $T/U \in \mathcal{L}(V/U)$, given by $(T/U)(v + U) = Tv + U$.

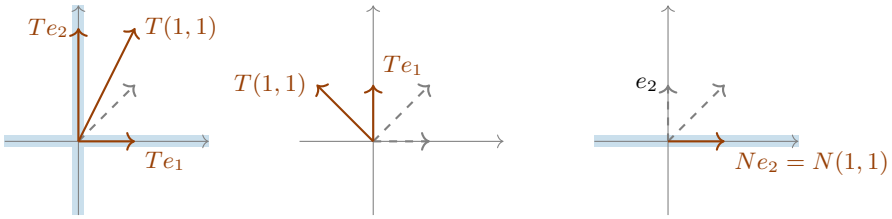
Proposition (Quotient operator)
If U is invariant under T, the formula $(T/U)(v + U) = Tv + U$ gives a well-defined operator on V/U.

Proof. "Well-defined" means the answer does not depend on which vector v we use to name the translate $v + U$. If $v + U = v' + U$, then $v - v' \in U$, so $Tv - Tv' = T(v - v') \in U$ because U is invariant. Hence $Tv + U = Tv' + U$. Linearity comes from linearity of T. □

Applying the formula k times gives $(T/U)^k(v + U) = T^k v + U$, so for every polynomial p (Section 1.11),

$$p(T/U)(v + U) = p(T)v + U, \qquad p(T|_U)u = p(T)u \quad (u \in U).$$

Example (Ex. 48)
$\{0\}$ and V are always invariant. $\operatorname{null} T$ is invariant: if $Tv = 0$ then $T(Tv) = T0 = 0$. $\operatorname{range} T$ is invariant: T of anything is in $\operatorname{range} T$.



$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$

Ex. 49. Dashed gray: some vectors. Solid: their images. Shaded: the invariant lines. Left: $\operatorname{diag}(1, 2)$ sends $e_1 \mapsto e_1$ and $e_2 \mapsto 2e_2$, so it keeps each axis, but it sends $(1, 1)$ to $(1, 2)$, off its line. Middle: rotation by 90° moves every line, so its only invariant subspaces are $\{0\}$ and $\mathbb{R}^2$. Right: N sends both e_2 and $(1, 1)$ to e_1, and $e_1 \mapsto 0$. Its only invariant line is the x-axis: if v is not on the x-axis, Nv is a nonzero vector on the x-axis, so not a multiple of v.

§1.11 Eigenvalues and eigenvectors

The idea. The smallest invariant subspaces are lines through 0. If the line span(v) is invariant, then Tv is a multiple of v: T only stretches v (or shrinks it, flips it, or sends it to 0).

Definition (Def. 31, 32 / LADR 5.5, 5.8, 5.52)**.** A scalar $\lambda \in \mathbb{F}$ is an **eigenvalue** of T if there is a vector $v \neq 0$ with $Tv = \lambda v$. Such a v is an **eigenvector** for λ. The **eigenspace** of λ is

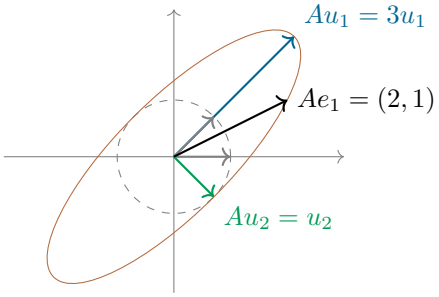
$$E(\lambda, T) = \operatorname{null}(T - \lambda I) = \{0\} \cup \{\text{eigenvectors for } \lambda\},$$

and its dimension is the **geometric multiplicity** of λ.

Why the null space: $Tv = \lambda v \iff Tv - \lambda v = 0 \iff (T - \lambda I)v = 0$. So

$$\lambda \text{ is an eigenvalue} \iff T - \lambda I \text{ is not injective} \iff T - \lambda I \text{ is not invertible},$$

where the last step uses that V is finite-dimensional. $E(\lambda, T)$ is invariant: if $Tv = \lambda v$ then $T(Tv) = \lambda Tv$.



$A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ sends the unit circle (dashed) to the ellipse. With $u_1 = \frac{1}{\sqrt{2}}(1, 1)$ and $u_2 = \frac{1}{\sqrt{2}}(1, -1)$ (unit vectors; the short gray arrows are u_1 and e_1, and u_2 is hidden under Au_2): along u_1 it stretches by 3; along u_2 it stretches by 1. These are the eigenvectors, with eigenvalues 3 and 1. The vector e_1 is not an eigenvector: $Ae_1 = (2, 1)$ points in a different direction.

Finding eigenvalues and eigenvectors.

Proposition (Prop. 56)

If $v_1, \ldots, v_n$ is a basis, then λ is an eigenvalue of T iff it is an eigenvalue of the matrix $\mathcal{M}(T)$, and v is an eigenvector of T iff its coordinate vector is an eigenvector of $\mathcal{M}(T)$. For a 2×2 matrix $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, the eigenvalues are the roots of the **characteristic polynomial**

$$\chi_A(t) = \det(tI - A) = t^2 - (a + d)\, t + (ad - bc).$$

Proof. $Tv = \lambda v$ is the same equation as $\mathcal{M}(T)\mathcal{M}(v) = \lambda \mathcal{M}(v)$ in coordinates. For 2×2: λ is an eigenvalue iff $\lambda I - A$ is not invertible iff $\det(\lambda I - A) = 0$, and $\det\left(\begin{smallmatrix} t-a & -b \\ -c & t-d \end{smallmatrix}\right) = (t - a)(t - d) - bc$. □

Example (Eigenvalues and eigenvectors of a 2×2 matrix)

$A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$: $\chi_A(t) = t^2 - 4t + 3 = (t - 1)(t - 3)$, so the eigenvalues are 1 and 3. For $\lambda = 3$: $A - 3I = \left(\begin{smallmatrix} -1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, whose null space is span$\big((1, 1)\big)$. For $\lambda = 1$: $A - I = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, whose null space is span$\big((1, -1)\big)$.

Example (Ex. 51: rotation)

$R = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ (rotation by 90°): $\chi_R(t) = t^2 + 1$. On $\mathbb{R}^2$ there are no eigenvalues. On $\mathbb{C}^2$ the eigenvalues are i and $-i$.

Polynomials applied to an operator. For $p(x) = a_m x^m + \cdots + a_1 x + a_0$, define (Def. 35)

$$p(T) = a_m T^m + \cdots + a_1 T + a_0 I.$$

For example, if $p(x) = x^2 - 3x + 2$ then $p(T) = T^2 - 3T + 2I$. The usual algebra works: $(p + q)(T) = p(T) + q(T)$ and $(pq)(T) = p(T)\, q(T)$. Since $T^j T^k = T^k T^j$, any two polynomials in T commute: $p(T)q(T) = q(T)p(T)$. For example $(T - I)(T - 2I) = T^2 - 3T + 2I = (T - 2I)(T - I)$.

Proposition (Eigenvalues through polynomials)

If $Tv = \lambda v$, then $p(T)v = p(\lambda)v$. Hence, if $p(T) = 0$, every eigenvalue of T is a root of p.

Proof. $T^2v = T(\lambda v) = \lambda^2 v$, and in the same way $T^k v = \lambda^k v$ for every k. Combine these with the coefficients of p. If $p(T) = 0$ and v is an eigenvector, then $0 = p(T)v = p(\lambda)v$ with $v \neq 0$, so $p(\lambda) = 0$. □

Example (Ex. 52)

$T\colon \mathbb{C}^{2,2} \to \mathbb{C}^{2,2}$, $T(A) = A^T$. Since $T^2 = I$, every eigenvalue satisfies $\lambda^2 = 1$, so $\lambda = \pm 1$. Both occur: symmetric matrices ($A^T = A$) form $E(1, T)$, of dimension 3, and $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ spans $E(-1, T)$. No determinant was needed.

More invariant subspaces: for every polynomial p, $\operatorname{null} p(T)$ and $\operatorname{range} p(T)$ are invariant under T, because T commutes with $p(T)$. For example, if $p(T)v = 0$ then $p(T)(Tv) = T\,p(T)v = 0$.

Theorem (Prop. 53 / LADR 5.19)

Every operator on a nonzero finite-dimensional *complex* vector space has an eigenvalue.

Proof. Let $n = \dim V$ and pick any $v \neq 0$. The $n + 1$ vectors $v, Tv, \ldots, T^n v$ are linearly dependent, so some combination $a_0 v + a_1 Tv + \cdots + a_m T^m v = 0$ with $a_m \neq 0$. Choose such a relation with m as small as possible and let $p(x) = a_0 + a_1 x + \cdots + a_m x^m$, so $p(T)v = 0$. Here $m \geq 1$, since $a_0 v = 0$ with $v \neq 0$ would force $a_0 = 0$. Over $\mathbb{C}$, p has a root λ, so $p(x) = (x - \lambda)q(x)$ with $\deg q = m - 1$. Put $w = q(T)v$. Then $w \neq 0$, because $q(T)v = 0$ would be a shorter relation. And $(T - \lambda I)w = (T - \lambda I)q(T)v = p(T)v = 0$. So w is an eigenvector with eigenvalue λ. □

Both hypotheses are needed. The rotation R on $\mathbb{R}^2$ has no eigenvalue (not complex). The operator $p \mapsto xp$ on $\mathcal{P}(\mathbb{C})$ has no eigenvector, since xp has larger degree than p (not finite-dimensional).

Proposition (Prop. 54, 55 / LADR 5.11, 5.12)

Eigenvectors for distinct eigenvalues are linearly independent. Hence T has at most $\dim V$ distinct eigenvalues.

Proof. Let $v_1, \ldots, v_m$ be eigenvectors with distinct eigenvalues $\lambda_1, \ldots, \lambda_m$, and suppose they are dependent. Let k be the smallest index with $v_k \in \operatorname{span}(v_1, \ldots, v_{k-1})$, and write $v_k = a_1 v_1 + \cdots + a_{k-1} v_{k-1}$. Apply $T - \lambda_k I$ to both sides. The left side becomes 0. The right side becomes $a_1(\lambda_1 - \lambda_k)v_1 + \cdots + a_{k-1}(\lambda_{k-1} - \lambda_k)v_{k-1}$. The vectors $v_1, \ldots, v_{k-1}$ are independent (by the choice of k) and each $\lambda_i - \lambda_k \neq 0$, so every $a_i = 0$. Then $v_k = 0$, but eigenvectors are nonzero. Contradiction. The second statement: m independent vectors in V means $m \leq \dim V$. □

Example (Ex. 50): $\sin x, \sin 2x, \ldots, \sin Nx$ are linearly independent, because they are eigenvectors of $f \mapsto f''$ with the distinct eigenvalues $-1, -4, \ldots, -N^2$.

Proposition (HW6 P3: similar matrices)

If S is invertible and $R = S^{-1}TS$, then T and R have the same eigenvalues, and $E(\lambda, R) = S^{-1}\big(E(\lambda, T)\big)$.

Proof. $Rv = \lambda v \iff S^{-1}TSv = \lambda v \iff T(Sv) = \lambda(Sv)$. So v is an eigenvector of R iff Sv is an eigenvector of T, with the same λ. □

Proposition (Eigenvalues of the dual map)

In finite dimensions, T and T' have the same eigenvalues, and $\dim E(\lambda, T) = \dim E(\lambda, T')$.

Proof. $(T - \lambda I)' = T' - \lambda I$ (the dual of the identity of V is the identity of V'). A map and its dual have the same rank, and V, V' have the same dimension, so by rank–nullity $T - \lambda I$ and $T' - \lambda I$ have null spaces of the same dimension. □

Stochastic matrices and PageRank (Prop. 57, 58). A **probability vector** has nonnegative entries that add up to 1. A matrix P is **(column-)stochastic** if its entries are nonnegative and each column adds up to 1. Think of a walker moving among n states (for example, web pages). If entry j of x is the probability of being at state j today, and P_{ij} is the probability of moving from state j to state i, then Px lists the probabilities for tomorrow. A **stationary distribution** is a probability vector π with $P\pi = \pi$: an eigenvector for eigenvalue 1, scaled so its entries add to 1. Example from the notes: $P = \begin{pmatrix} 1/2 & 1/4 & 0 \\ 1/2 & 1/2 & 1/2 \\ 0 & 1/4 & 1/2 \end{pmatrix}$ has $\pi = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$.

Proposition (Prop. 57)
Every eigenvalue λ of a stochastic matrix P satisfies $|\lambda| \leq 1$.

Proof. Write $\|x\|_1 = |x_1| + \cdots + |x_n|$. Since the entries are nonnegative and each column sums to 1, $\|Px\|_1 \leq \sum_i \sum_j P_{ij}|x_j| = \sum_j |x_j| = \|x\|_1$. If $Px = \lambda x$ with $x \neq 0$, then $|\lambda|\,\|x\|_1 = \|Px\|_1 \leq \|x\|_1$, so $|\lambda| \leq 1$. □

PageRank changes the walk: at each step, with probability α the walker follows a link (moves by P), and otherwise jumps to a page chosen with the probabilities in a fixed probability vector v. One step sends x to $\alpha Px + (1-\alpha)v$, and the PageRank vector r is the probability vector that this step leaves unchanged.

Proposition (Prop. 58: PageRank exists and is unique)
Let P be stochastic, v a probability vector, and $0 < \alpha < 1$. Then $r = \alpha Pr + (1-\alpha)v$ has exactly one solution r, and r is a probability vector.

Proof outline. The equation is $(I - \alpha P)r = (1-\alpha)v$. If $(I-\alpha P)x = 0$ with $x \neq 0$, then $Px = \frac{1}{\alpha}x$, so $\frac{1}{\alpha} > 1$ would be an eigenvalue, contradicting Prop. 57. So $I - \alpha P$ is injective, hence invertible, and the solution exists and is unique. It equals $(1-\alpha)\sum_{k\geq 0} \alpha^k P^k v$: multiplying out, $(I-\alpha P)\sum_{k=0}^{K} \alpha^k P^k v = v - \alpha^{K+1}P^{K+1}v$ because the middle terms cancel, and $\alpha^{K+1}P^{K+1}v \to 0$. This series converges, since each $P^k v$ is a probability vector and $\sum \alpha^k < \infty$. Every term is nonnegative, and the entries of the sum add up to $(1-\alpha)\sum_k \alpha^k = 1$. So r is a probability vector. □

The Google matrix. Let $\mathbf{1} = (1, \ldots, 1)$ and $G = \alpha P + (1-\alpha)v\mathbf{1}^T$. Since $\mathbf{1}^T x = 1$ for a probability vector x, $Gx = \alpha Px + (1-\alpha)v$, so the PageRank equation is the eigenvector equation $Gr = r$. The subspace $H = \{x : x_1 + \cdots + x_n = 0\}$ is invariant under P (which preserves the sum of the entries) and under G, where $Gx = \alpha Px$ for $x \in H$. Since $\mathbf{1}^T r = 1 \neq 0$, $r \notin H$, so $\mathbb{R}^n = \mathrm{span}(r) \oplus H$: G fixes the PageRank direction, and on H it acts as αP, so every eigenvalue of G coming from H has absolute value at most α (Prop. 57).

The iteration $r_{k+1} = \alpha Pr_k + (1-\alpha)v$ converges to r from any starting probability vector r_0: $\|r_k - r\|_1 \leq \alpha^k \|r_0 - r\|_1$, because $r_{k+1} - r = \alpha P(r_k - r)$ and P does not increase $\|\cdot\|_1$.

Traps.

- Eigenvectors are nonzero by definition. Eigenvalues may be 0: $\lambda = 0$ is an eigenvalue iff T is not injective.
- "$p(T) = 0$ and $p(\lambda) = 0$" does *not* make λ an eigenvalue. $T = I$ satisfies $(T-I)(T-2I) = 0$, but 2 is not an eigenvalue of I.
- Eigenvalues depend on the field: the rotation R has none over $\mathbb{R}$ and two over $\mathbb{C}$.

§1.12 Upper-triangular matrices

The idea. Many operators (for example the shear $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$) have no basis of eigenvectors. But over $\mathbb{C}$ every operator has a basis in which its matrix is upper-triangular, and that is almost as useful: the eigenvalues can still be read off the diagonal.

Definition (Def. 33)**.** A square matrix $A = (a_{ij})$ is **upper-triangular** if every entry below the diagonal is 0: $a_{ij} = 0$ whenever $i > j$. The **diagonal** is $a_{11}, \ldots, a_{nn}$. A is **diagonal** if every entry off the diagonal is 0.

Example (Ex. 53): $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 5 & 6 & 7 \\ 0 & 0 & 8 & 9 \\ 0 & 0 & 0 & 10 \end{pmatrix}$ is upper-triangular with diagonal $1, 5, 8, 10$.

What upper-triangular means for T. Column j of $\mathcal{M}(T)$ holds the coefficients of Tv_j. Zeros below the diagonal in column j mean Tv_j uses only $v_1, \dots, v_j$:

$$\mathcal{M}(T) = \begin{pmatrix} \lambda_1 & * & \cdots & * \\ 0 & \lambda_2 & \cdots & * \\ \vdots & & \ddots & \vdots \\ 0 & \cdots & 0 & \lambda_n \end{pmatrix}, \qquad \begin{aligned} Tv_1 &= \lambda_1 v_1, \\ Tv_2 &= * v_1 + \lambda_2 v_2, \\ &\vdots \\ Tv_n &= * v_1 + \cdots + * v_{n-1} + \lambda_n v_n. \end{aligned}$$

So Tv_j is $\lambda_j v_j$ plus a combination of $v_1, \dots, v_{j-1}$; no later basis vector appears. In particular v_1 is an eigenvector.

Proposition (Prop. 59 / LADR 5.39)

For a basis $v_1, \dots, v_n$ of V, these are equivalent:

1. $\mathcal{M}(T)$ is upper-triangular;
2. $Tv_j \in \operatorname{span}(v_1, \dots, v_j)$ for each j;
3. $\operatorname{span}(v_1, \dots, v_j)$ is invariant under T for each j.

Proof. (1)⇔(2) is the column description above. (2)⇒(3): for $i \le j$, $Tv_i \in \operatorname{span}(v_1, \dots, v_i) \subseteq \operatorname{span}(v_1, \dots, v_j)$, so T maps the spanning list of $\operatorname{span}(v_1, \dots, v_j)$ into it. (3)⇒(2): $v_j \in \operatorname{span}(v_1, \dots, v_j)$ and that span is invariant. □

So an upper-triangular matrix is the same as a growing chain of invariant subspaces $\operatorname{span}(v_1) \subseteq \operatorname{span}(v_1, v_2) \subseteq \cdots \subseteq V$, gaining one dimension at each step.

Proposition (Prop. 60 / LADR 5.41)

If $\mathcal{M}(T)$ is upper-triangular, the eigenvalues of T are exactly its diagonal entries. In particular, T is invertible iff no diagonal entry is 0.

Proof. $\mathcal{M}(T - \lambda I)$ is upper-triangular with diagonal $\lambda_1 - \lambda, \dots, \lambda_n - \lambda$. If λ is not on the diagonal, all these are nonzero. Solve $(T - \lambda I)x = 0$ from the bottom row up: the last row says $(\lambda_n - \lambda)x_n = 0$, so $x_n = 0$; then the next row gives $x_{n-1} = 0$; and so on ("back-substitution"). So $T - \lambda I$ is injective, and λ is not an eigenvalue. If $\lambda = \lambda_k$, the first k columns of $\mathcal{M}(T - \lambda I)$ have zeros in rows $k, \dots, n$, so they are k vectors in a space of dimension $k - 1$. They are dependent, so $T - \lambda I$ is not injective. □

Example (Ex. 54)

Differentiation D on $\mathcal{P}_3(\mathbb{R})$ in the basis $1, x, x^2, x^3$ has matrix

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

It is upper-triangular with diagonal $0, 0, 0, 0$, so the only eigenvalue is 0. In words: $x^3 \mapsto 3x^2 \mapsto 6x \mapsto 6 \mapsto 0$, each basis vector sent to a multiple of the previous one.

Theorem (Prop. 61 / LADR 5.47)

If V is a finite-dimensional complex vector space, every $T \in \mathcal{L}(V)$ has an upper-triangular matrix in some basis.

Proof. Induction on $\dim V$; dimension 1 is clear. Take an eigenvalue λ (Prop. 53) and let $U = \operatorname{range}(T - \lambda I)$.

- U is invariant: $T(T - \lambda I)v = (T - \lambda I)(Tv) \in U$.
- $\dim U < \dim V$: $T - \lambda I$ is not injective, so it is not surjective.

By induction, U has a basis $u_1, \dots, u_m$ in which $T|_U$ is upper-triangular. Extend it to a basis $u_1, \dots, u_m, v_1, \dots, v_k$ of V. Then

$$Tv_j = \underbrace{(T - \lambda I)v_j}_{\in U} + \lambda v_j \in \operatorname{span}(u_1, \dots, u_m, v_j),$$

so condition (2) of Prop. 59 holds for the v_j. It holds for the u_i because $T|_U$ is upper-triangular in $u_1, \dots, u_m$. □

Example (Triangularizing a matrix that cannot be diagonalized)
$A = \left(\begin{smallmatrix} 4 & 1 \\ -1 & 2 \end{smallmatrix}\right)$: $\chi_A(t) = t^2 - 6t + 9 = (t-3)^2$, and $A - 3I = \left(\begin{smallmatrix} 1 & 1 \\ -1 & -1 \end{smallmatrix}\right)$ has null space span $\big((1,-1)\big)$. Take $s_1 = (1,-1)$ and extend by $s_2 = e_1$. Then $As_1 = 3s_1$ and $As_2 = (4,-1) = 1 \cdot s_1 + 3 \cdot s_2$. So in the basis s_1, s_2 the matrix is $\left(\begin{smallmatrix} 3 & 1 \\ 0 & 3 \end{smallmatrix}\right)$. (Equivalently $S^{-1}AS$ with $S = \left(\begin{smallmatrix} 1 & 1 \\ -1 & 0 \end{smallmatrix}\right)$.)

Proposition (LADR 5.40)
If $\mathcal{M}(T)$ is upper-triangular with diagonal $\lambda_1, \dots, \lambda_n$, then $(T - \lambda_1 I)(T - \lambda_2 I) \cdots (T - \lambda_n I) = 0$.

Proof. Upper-triangular means $(T - \lambda_k I)v_k \in \operatorname{span}(v_1, \dots, v_{k-1})$. We show by induction on k that $(T - \lambda_1 I) \cdots (T - \lambda_k I)$ sends $v_1, \dots, v_k$ to 0. The factors commute, so apply $T - \lambda_k I$ first: it sends v_k into $\operatorname{span}(v_1, \dots, v_{k-1})$, which the remaining product sends to 0 by induction; and $v_1, \dots, v_{k-1}$ are already sent to 0 by the remaining product. At $k = n$ the product sends a basis to 0. □

Over $\mathbb{R}$, which operators can be made upper-triangular is answered in Section 1.14 (LADR 5.44).

Traps.

- A triangular matrix (upper or lower) shows its eigenvalues on the diagonal. For a general matrix the diagonal entries need not be eigenvalues: $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ has zeros on the diagonal and eigenvalues ± 1.
- Over $\mathbb{R}$ upper-triangular form can fail: the rotation R has no eigenvalue, so no first basis vector v_1 can be an eigenvector.
- Upper-triangular is not unique: different bases give different entries above the diagonal, and the diagonal entries can appear in a different order.

§1.13 Diagonalizability

The idea. The best case is a basis made of eigenvectors. In that basis the matrix is diagonal: T just multiplies each coordinate by its own number. Then everything is easy to compute, especially powers: T^k multiplies each coordinate by λ^k.

Definition (Def. 34 / LADR 5.50)**.** T is **diagonalizable** if $\mathcal{M}(T)$ is diagonal for some basis of V. A basis of V consisting of eigenvectors of T is an **eigenbasis**.



diag$(2, \frac{1}{2})$ shear $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$

What each matrix does to the unit square (dashed). Left: diagonal, so the square is stretched along the two eigenvector directions e_1 and e_2. Right: the shear slides each horizontal line sideways by its height. Only vectors on the x-axis keep their direction, so there is just one line of eigenvectors, too few for an eigenbasis of $\mathbb{R}^2$. The shear is not diagonalizable.

Theorem (Prop. 62 / LADR 5.55)

Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of T. These are equivalent:

1. T is diagonalizable;
2. V has a basis of eigenvectors of T;
3. $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$;
4. $\dim V = \dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T)$.

Proof. (1)⇔(2): a matrix is diagonal exactly when each basis vector is sent to a multiple of itself, that is, each basis vector is an eigenvector. The sum $E(\lambda_1, T) + \cdots + E(\lambda_m, T)$ is always direct: a vector $0 = w_1 + \cdots + w_m$ with $w_i \in E(\lambda_i, T)$ forces every $w_i = 0$, because nonzero eigenvectors for distinct eigenvalues are independent (Prop. 54). (2)⇒(3): the eigenbasis lies in the sum, so the sum is all of V. (3)⇒(4): dimensions add in a direct sum. (4)⇒(3): the direct sum has dimension $\dim V$, so it is V. (3)⇒(2): put together a basis of each eigenspace. □

The practical test is (4). For each eigenvalue compute $\dim E(\lambda, T) = \dim V - \operatorname{rank}(T - \lambda I)$, add them, and compare with $\dim V$.

Example (Diagonalizing a matrix)

$A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ has eigenvectors $(1, 1)$ for 3 and $(1, -1)$ for 1. Put them as the columns of P:

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad D = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}, \qquad A = PDP^{-1}.$$

(D is the matrix of the same operator in the eigenbasis; this is the change-of-basis formula in Section 1.1.) Then $A^k = PD^kP^{-1}$, which works out to $\frac{1}{2}\left(\begin{smallmatrix} 3^k+1 & 3^k-1 \\ 3^k-1 & 3^k+1 \end{smallmatrix}\right)$.

Example (Not diagonalizable)

$A = \left(\begin{smallmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 3 \end{smallmatrix}\right)$ is upper-triangular, so its eigenvalues are 1 and 3. $\operatorname{rank}(A - I) = 2$ and $\operatorname{rank}(A - 3I) = 2$, so $\dim E(1, A) = 1$ and $\dim E(3, A) = 1$. Since $1 + 1 = 2 < 3$, A is not diagonalizable.

Proposition (Prop. 63 / LADR 5.58)

If T has $\dim V$ distinct eigenvalues, it is diagonalizable.

Proof. One eigenvector for each eigenvalue gives $\dim V$ independent vectors (Prop. 54), which is a basis. □

The converse is false: I is diagonal and has only one eigenvalue.

Example (Ex. 55): for $n \geq 2$, the shift $Tv_1 = 0$, $Tv_k = v_{k-1}$ has upper-triangular matrix with zero diagonal, so its only eigenvalue is 0, and $\dim E(0, T) = \dim \operatorname{null} T = 1 < n$. So it is not diagonalizable: every space of dimension at least 2 has non-diagonalizable operators.

Diagonalizability depends on the field (Ex. 56). $\left(\begin{smallmatrix} 1 & 2 \\ 2 & 3 \end{smallmatrix}\right)$ has the distinct eigenvalues $2 \pm \sqrt{5}$, so it is diagonalizable over $\mathbb{R}$, but over the rational numbers $\mathbb{Q}$ it has no eigenvalues at all. The rotation R is diagonalizable over $\mathbb{C}$ (eigenvalues $\pm i$, distinct) but not over $\mathbb{R}$ (no eigenvalues at all).

Proposition (Prop. 64: powers in an eigenbasis)

If $Tv_j = \lambda_j v_j$ for a basis $v_1, \ldots, v_n$ and $v = a_1 v_1 + \cdots + a_n v_n$, then for every $k \geq 0$

$$T^k v = a_1 \lambda_1^k v_1 + \cdots + a_n \lambda_n^k v_n.$$

For matrices: if $A = PDP^{-1}$ with D diagonal, then $A^k = PD^kP^{-1}$, and D^k is D with each diagonal entry raised to the kth power.

Proof. $T^k v_j = \lambda_j^k v_j$; apply linearity. For matrices, in $(PDP^{-1})(PDP^{-1}) \cdots (PDP^{-1})$ each inner $P^{-1}P$ cancels, leaving PD^kP^{-1}. □

Example (A Markov chain from the notes)

$P = \begin{pmatrix} 1/2 & 1/4 & 0 \\ 1/2 & 1/2 & 1/2 \\ 0 & 1/4 & 1/2 \end{pmatrix}$ has the eigenbasis

$$\pi = \begin{pmatrix} 1/4 \\ 1/2 \\ 1/4 \end{pmatrix} \ (\lambda = 1), \qquad u = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \ (\lambda = \tfrac{1}{2}), \qquad w = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \ (\lambda = 0).$$

The starting distribution $p_0 = (1, 0, 0)$ is $\pi - \frac{1}{2}u + \frac{1}{4}w$. By Prop. 64, for $k \geq 1$,

$$P^k p_0 = \pi - \tfrac{1}{2}\left(\tfrac{1}{2}\right)^k u + 0 = \pi - 2^{-(k+1)}u.$$

The w part disappears after one step, the u part halves at every step, and $P^k p_0 \to \pi$.

Example (Spectral gap)

For $0 < \varepsilon < \frac{1}{2}$, $P_\varepsilon = \left(\begin{smallmatrix} 1-\varepsilon & \varepsilon \\ \varepsilon & 1-\varepsilon \end{smallmatrix}\right)$ has eigenvectors $(1, 1)$ for eigenvalue 1 and $(1, -1)$ for $1 - 2\varepsilon$, so $P_\varepsilon^k(1, -1) = (1 - 2\varepsilon)^k(1, -1)$. When ε is small (the walk rarely switches states), $1 - 2\varepsilon$ is close to 1 and the starting state is forgotten slowly. The difference $1 - (1 - 2\varepsilon) = 2\varepsilon$ is called the **spectral gap**.

Proposition (HW6 P1: projections)

If $P^2 = P$, then every eigenvalue of P is 0 or 1, $V = \operatorname{null} P \oplus \operatorname{range} P$, and $\operatorname{range} P = E(1, P)$. So P is diagonalizable.

Proof. Any v is $(v - Pv) + Pv$, with $P(v - Pv) = Pv - P^2v = 0$ and $P(Pv) = Pv$. So $V = \operatorname{null} P + E(1, P)$, and $\operatorname{range} P \subseteq E(1, P) \subseteq \operatorname{range} P$. The sum is direct: if $v = Pw$ and $Pv = 0$, then $v = Pw = P^2w = Pv = 0$. □

Traps.

- Repeated eigenvalues do not rule out diagonalizability (I is diagonal); they only mean you must check the dimensions in test (4).
- "Diagonalizable" and "invertible" are unrelated. The zero operator is diagonalizable and not invertible. The shear $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ is invertible and not diagonalizable.
- In $A = PDP^{-1}$ the eigenvectors are the *columns* of P, in the same order as the eigenvalues in D.

§1.14 The minimal polynomial

The idea. If $\dim V = n$, then $\mathcal{L}(V)$ has dimension n^2, so the list $I, T, T^2, T^3, \ldots$ must become linearly dependent. A dependence $a_0 I + a_1 T + \cdots + a_k T^k = 0$ says $p(T) = 0$ for the polynomial $p = a_0 + a_1 x + \cdots + a_k x^k$. The shortest such equation is the minimal polynomial. Its roots are exactly the eigenvalues, and whether it has repeated roots decides whether T is diagonalizable.

Definition (Def. 36 / LADR 5.22, 5.24)**.** Let k be the smallest number such that $T^k \in \operatorname{span}(I, T, \ldots, T^{k-1})$, and write $T^k = a_{k-1}T^{k-1} + \cdots + a_1 T + a_0 I$. The **minimal polynomial** of T is

$$m_T(x) = x^k - a_{k-1}x^{k-1} - \cdots - a_1 x - a_0.$$

Equivalently: m_T is monic, $m_T(T) = 0$, and no nonzero polynomial q of smaller degree has $q(T) = 0$.

Example (Ex. 57, 58)

- On $V \neq \{0\}$ (of any finite dimension): $m_I = x - 1$, since $I - I = 0$; and $m_0 = x$.
- $T = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$. I and T are independent (T is not a multiple of I). Compute $T^2 = \left(\begin{smallmatrix} 5 & 4 \\ 4 & 5 \end{smallmatrix}\right)$ and solve $T^2 = aT + bI$: the off-diagonal entries give $4 = a$, the diagonal gives $5 = 2a + b$, so $b = -3$. Hence $T^2 = 4T - 3I$ and $m_T = x^2 - 4x + 3 = (x-1)(x-3)$.
- Rotation: $R^2 = -I$ and R is not a multiple of I, so $m_R = x^2 + 1$.
- The $n \times n$ shift N ($Ne_1 = 0$, $Ne_j = e_{j-1}$) has $N^n = 0$, so (by Prop. 65 below) m_N divides x^n, that is, $m_N = x^j$ with $j \leq n$. Since $N^{n-1} \neq 0$, $j = n$: $m_N = x^n$.

Proposition (Prop. 65 / LADR 5.29)

For a polynomial q: $q(T) = 0$ iff m_T divides q.

Proof. If $q = m_T r$, then $q(T) = m_T(T)r(T) = 0$. Conversely, suppose $q(T) = 0$. Divide with remainder: $q = m_T r + s$ with $s = 0$ or $\deg s < \deg m_T$. Then $s(T) = q(T) - m_T(T)r(T) = 0$. A nonzero s would be a polynomial of smaller degree than m_T with $s(T) = 0$, which is impossible. So $s = 0$ and m_T divides q. □

So the minimal polynomial is a factor of *every* polynomial q with $q(T) = 0$. If you find one such polynomial, m_T is among its monic factors.

Proposition (Prop. 66 / LADR 5.27)

The eigenvalues of T are exactly the roots of m_T that lie in $\mathbb{F}$.

Proof. If $Tv = \lambda v$ with $v \neq 0$, then $0 = m_T(T)v = m_T(\lambda)v$, so $m_T(\lambda) = 0$. Conversely, let $m_T(\lambda) = 0$, so $m_T = (x - \lambda)q$ with $\deg q < \deg m_T$. If λ were not an eigenvalue, $T - \lambda I$ would be invertible, and $0 = m_T(T) = (T - \lambda I)q(T)$ would give $q(T) = 0$, contradicting minimality. □

Proposition (Prop. 67 / LADR 5.22)

$\deg m_T \leq \dim V$.

Proof outline. Induction on $\dim V$. Pick $v \neq 0$; the $n + 1$ vectors $v, \ldots, T^n v$ are dependent, so $q(T)v = 0$ for some nonzero q of degree at most n. If $q(T) = 0$, then $\deg m_T \leq \deg q \leq n$. Otherwise $U = \operatorname{null} q(T)$ is invariant, contains v, and is not V. By induction $\deg m_{T|_U} \leq \dim U$ and $\deg m_{T/U} \leq \dim V/U$. The product $r = m_{T|_U} m_{T/U}$ has $r(T) = 0$: since $m_{T/U}(T/U) = 0$, the formula $p(T/U)(v + U) = p(T)v + U$ (Section 1.10) gives $m_{T/U}(T)v \in U$ for every v, and then $m_{T|_U}(T)$ sends that vector of U to 0. So m_T divides r (Prop. 65), and $\deg m_T \leq \deg r \leq \dim U + \dim V/U = \dim V$. □

Theorem (Prop. 68 / LADR 5.62)

T is diagonalizable iff $m_T = (x - \lambda_1) \cdots (x - \lambda_m)$ with $\lambda_1, \ldots, \lambda_m \in \mathbb{F}$ all different ("m_T splits with no repeated root").

Proof. ($\Rightarrow$) Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues, so $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$. The product $(T - \lambda_1 I) \cdots (T - \lambda_m I)$ sends each $E(\lambda_i, T)$ to 0 (the factors commute, so apply $T - \lambda_i I$ first), hence sends all of V to 0. So m_T divides $(x - \lambda_1) \cdots (x - \lambda_m)$, and it has every λ_i as a root (Prop. 66), so it equals this product.

($\Leftarrow$) Induction on $\dim V$. If $m = 1$, $T = \lambda_1 I$. Otherwise, the key step is

$$V = \operatorname{null}(T - \lambda_1 I) \oplus \operatorname{range}(T - \lambda_1 I).$$

Divide $g(x) = (x - \lambda_2) \cdots (x - \lambda_m)$ by $x - \lambda_1$: $g = (x - \lambda_1)q + s$ with s a constant, and $s = g(\lambda_1) \neq 0$ since the roots are different. For $v \in V$ let $u = \frac{1}{s} g(T)v$. Then $(T - \lambda_1 I)u = \frac{1}{s} m_T(T)v = 0$, so $u \in \operatorname{null}(T - \lambda_1 I)$, and $v - u = -\frac{1}{s}(T - \lambda_1 I)q(T)v \in \operatorname{range}(T - \lambda_1 I)$. So V is the sum of the two spaces, and their dimensions add up to

$\dim V$ by rank–nullity, so the sum is direct (Section 1.1). The range $U = \text{range}(T - \lambda_1 I)$ is invariant, and smaller than V because λ_1 is a root of m_T, hence an eigenvalue (Prop. 66), so $\text{null}(T - \lambda_1 I) \neq \{0\}$. Also $m_{T|_U}$ divides m_T (item 3 of the corollary below: $m_T(T|_U) = 0$, then Prop. 65), so it also has distinct roots; by induction $T|_U$ has an eigenbasis. Together with a basis of $\text{null}(T - \lambda_1 I)$, this is an eigenbasis of V. □

Over $\mathbb{C}$ you can test "no repeated root" without finding the roots: p has a repeated root iff p and p' have a common root, and that can be checked by computing the greatest common divisor of p and p' with repeated division with remainder (the Euclidean algorithm): there is a common root iff this divisor is not a constant.

Proposition (LADR 5.44)
Over any $\mathbb{F}$: T has an upper-triangular matrix in some basis iff m_T splits over $\mathbb{F}$.

Proof. ($\Rightarrow$) If $\mathcal{M}(T)$ is upper-triangular with diagonal $\lambda_1, \ldots, \lambda_n$, then $(T - \lambda_1 I) \cdots (T - \lambda_n I) = 0$ (LADR 5.40), so by Prop. 65 m_T divides this product and splits. ($\Leftarrow$) Repeat the proof of Prop. 61. An eigenvalue exists because m_T splits and its roots are eigenvalues (Prop. 66). For the invariant $U = \text{range}(T - \lambda I)$, $m_{T|_U}$ divides m_T (see below), so it also splits and the induction applies to $T|_U$. □

So over $\mathbb{R}$, an operator can be made upper-triangular exactly when m_T has no irreducible quadratic factor.

Corollary (Consequences used often)
1. (Ex. 59) Over $\mathbb{C}$, if $T^n = I$ then T is diagonalizable: m_T divides $x^n - 1$, whose roots are distinct (it shares no root with nx^{n-1}).
2. If $p(T) = 0$ for some p that splits with no repeated root, then T is diagonalizable: m_T divides p, so m_T also splits with no repeated root. You do not need p to be the minimal polynomial.
3. If U is invariant, then $m_{T|_U}$ and $m_{T/U}$ divide m_T. Reason: $m_T(T) = 0$ gives $m_T(T|_U) = 0$ and $m_T(T/U) = 0$ (by $p(T/U)(v + U) = p(T)v + U$), so Prop. 65 applies. (This is item (1) of technique T5, Section 2.5.)
4. (Ex. 60) If T is diagonalizable and U is invariant, then $T|_U$ and T/U are diagonalizable, because $m_{T|_U}$ and $m_{T/U}$ divide m_T.
5. (HW7 P3) T is invertible iff $m_T(0) \neq 0$ (iff 0 is not an eigenvalue). Then T^{-1} is a polynomial in T: writing $m_T = x^d + a_{d-1}x^{d-1} + \cdots + a_1 x + a_0$ (so $a_0 = m_T(0) \neq 0$), from $m_T(T) = 0$:
$$T^{-1} = -\tfrac{1}{a_0}\big(T^{d-1} + a_{d-1}T^{d-2} + \cdots + a_2 T + a_1 I\big).$$
6. (HW7 P4) $m_{T-\lambda I}(x) = m_T(x + \lambda)$, and for $c \neq 0$, $m_{cT}(x) = c^d m_T(x/c)$ where $d = \deg m_T$.
7. (HW7 P5) A 2×2 complex matrix A whose only eigenvalue is λ has $m_A = x - \lambda$ (and then $A = \lambda I$) or $m_A = (x - \lambda)^2$ (and then A is not diagonalizable).

How to compute m_T.

- When m_T splits (always over $\mathbb{C}$): $m_T = \prod_i (x - \lambda_i)^{k_i}$ over the distinct eigenvalues, with each $k_i \geq 1$ and $\sum k_i \leq \dim V$. Test candidate products in order of increasing degree; the first one that gives 0 is m_T.
- When it may not split (over $\mathbb{R}$): use the definition (find the first k with $T^k \in \text{span}(I, \ldots, T^{k-1})$). Or, if for some v the list $v, Tv, \ldots, T^{n-1}v$ is a basis ($n = \dim V$; such v is called a cyclic vector; see technique T6, Section 2.6), write $T^n v = c_0 v + \cdots + c_{n-1}T^{n-1}v$; then $m_T = x^n - c_{n-1}x^{n-1} - \cdots - c_0$ (Practice Midterm C, Q3).

Example (Computing m_T for a 3×3 matrix)
$A = \left(\begin{smallmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 3 \end{smallmatrix}\right)$. Eigenvalues (diagonal): $1, 3$. So $(x - 1)(x - 3)$ divides m_A. Test it:

$$(A - I)(A - 3I) = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} -2 & 1 & 1 \\ 0 & -2 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \neq 0.$$

Next test $(x - 1)^2(x - 3)$: multiply the result by $A - I$ on the left to get 0. So m_A divides $(x - 1)^2(x - 3)$, has both roots, and is not $(x - 1)(x - 3)$; hence $m_A = (x - 1)^2(x - 3)$. It has a repeated root, so A is not diagonalizable (matching the dimension count in Section 1.13).

Proposition (Real operators)

Let V be a nonzero finite-dimensional real vector space and $T \in \mathcal{L}(V)$.

1. T has an invariant subspace of dimension 1 or 2.
2. If $\dim V$ is odd, T has an eigenvalue.

Proof. (1) Factor m_T over $\mathbb{R}$ into degree-1 factors and irreducible quadratics. If there is a degree-1 factor $x - \lambda$, then λ is an eigenvalue (Prop. 66) and an eigenvector spans an invariant line. Otherwise write $m_T = (x^2 + bx + c)q$. By minimality $q(T) \neq 0$, so pick v with $w = q(T)v \neq 0$. Then $(T^2 + bT + cI)w = m_T(T)v = 0$, so $T(Tw) = -bTw - cw$, and $\operatorname{span}(w, Tw)$ is invariant, of dimension 1 or 2. (2) Induction on $\dim V$; dimension 1 is clear. Suppose $\dim V \geq 3$ is odd and T has no eigenvalue. By (1) there is an invariant U of dimension 1 or 2; dimension 1 would give an eigenvector, so $\dim U = 2$. Then $\dim V/U = \dim V - 2$ is odd and smaller, so by induction T/U has an eigenvalue λ, which is a root of $m_{T/U}$. Since $m_{T/U}$ divides m_T, λ is a root of m_T, hence an eigenvalue of T (Prop. 66). Contradiction. □

Traps.

- Do not confuse m_A with the 2×2 characteristic polynomial $\chi_A(t) = t^2 - (a + d)t + (ad - bc)$. For a 2×2 matrix, $A^2 - (a + d)A + (ad - bc)I = 0$ (multiply it out), so m_A divides χ_A; and $m_A = \chi_A$ unless $A = \lambda I$, in which case $m_A = x - \lambda$.
- Over $\mathbb{R}$, m_T can have an irreducible quadratic factor ($x^2 + 1$ for the rotation). Then T is not diagonalizable over $\mathbb{R}$, even if the roots in $\mathbb{C}$ are distinct.
- A root of some p with $p(T) = 0$ need not be an eigenvalue; a root of m_T always is.

§2 Techniques

§2.1 A polynomial equation limits the eigenvalues

Theorem (T1)

Suppose $p(T) = 0$ for a nonzero polynomial p.

1. Every eigenvalue of T is a root of p.
2. If $V \neq \{0\}$ and $p = c(x - \mu_1)\cdots(x - \mu_k)$ splits over $\mathbb{F}$, then at least one μ_i is an eigenvalue.
3. A root μ of p that is not an eigenvalue can be removed: if $p = (x - \mu)q$ and μ is not an eigenvalue, then $q(T) = 0$.

Why. (1) If $Tv = \lambda v$ with $v \neq 0$, then $0 = p(T)v = p(\lambda)v$, so $p(\lambda) = 0$. (2) The product $(T - \mu_1 I)\cdots(T - \mu_k I)$ equals $\frac{1}{c}p(T) = 0$, which is not injective on $V \neq \{0\}$. A product of injective maps is injective, so at least one factor $T - \mu_i I$ is not injective, and μ_i is an eigenvalue. (3) For every v, $(T - \mu I)\big(q(T)v\big) = p(T)v = 0$. Since μ is not an eigenvalue, $T - \mu I$ is injective, so $q(T)v = 0$.

Past Berkeley exams ask this as $T^3 = T$, $(T - aI)(T - bI) = 0$, $T^2 = T$, $T^4 = -T^2$.

Problems.

(a) V is a nonzero real vector space and $T^3 = T$. Prove that the eigenvalues of T lie in $\{0, 1, -1\}$ and that at least one of them is an eigenvalue.

(b) $(T - aI)(T - bI) = 0$ and T is not a scalar multiple of I. Prove that a and b are both eigenvalues. (Ribet, Spring 2014.)

(c) $T \in \mathcal{L}(\mathbb{C}^2)$ satisfies $T^4 = -T^2$. Which values can be eigenvalues? Can one T have all of them? Give a T for each pair that can occur together.

(d) Is this true: "if $f(T) = 0$ for a real polynomial f of odd degree, then T has a real eigenvalue"? (Ribet, Spring 2014.)

§2.2 Distinct roots give diagonalizability, with explicit formulas

Theorem (T2)

If $p(T) = 0$ where $p = (x - \lambda_1)\cdots(x - \lambda_m)$ with $\lambda_1, \ldots, \lambda_m \in \mathbb{F}$ all different, then T is diagonalizable. Explicitly, define the **Lagrange polynomials**

$$p_j(x) = \prod_{k \neq j} \frac{x - \lambda_k}{\lambda_j - \lambda_k} \qquad (j = 1, \ldots, m).$$

Then every v can be written as $v = \sum_j p_j(T)v$, with $p_j(T)v \in E(\lambda_j, T)$.

Why. Each p_j has degree $m - 1$, equals 1 at λ_j, and equals 0 at every other λ_k. So $p_1 + \cdots + p_m - 1$ is a polynomial of degree less than m with the m roots $\lambda_1, \ldots, \lambda_m$, which forces it to be the zero polynomial: $\sum_j p_j = 1$, hence $\sum_j p_j(T) = I$ and $v = \sum_j p_j(T)v$. Also $(x - \lambda_j)p_j(x)$ is a constant times $p(x)$, so $(T - \lambda_j I)p_j(T)v = 0$, which says $p_j(T)v \in E(\lambda_j, T)$. So every vector is a sum of eigenvectors. (HW7 P7.)

When a problem only asks whether T is diagonalizable, cite LADR 5.62 (Prop. 68). Use the formula when a problem asks you to *write out* the decomposition.

Problems.

(a) $T^2 = T + 2I$. Prove T is diagonalizable and write v explicitly as a sum of eigenvectors.

(b) $T^3 = T$ on a real space. Prove $V = \operatorname{null} T \oplus E(1, T) \oplus E(-1, T)$, with explicit formulas for the three components of v.

(c) Prove that over $\mathbb{C}$, $T^n = I$ implies T diagonalizable. Give a real operator with $T^4 = I$ that is not diagonalizable over $\mathbb{R}$.

(d) Suppose $p(T) = 0$ with p having distinct roots in $\mathbb{F}$, and U is T-invariant. Prove $T|_U$ is diagonalizable in one line.

§2.3 Commuting operators share invariant subspaces

Theorem (T3)

If $ST = TS$, then S maps each of these subspaces into itself: $E(\lambda, T)$, $\operatorname{null} p(T)$, $\operatorname{range} p(T)$. Consequently, on a nonzero finite-dimensional complex space, commuting operators have a common eigenvector. Two commuting diagonalizable operators (over any $\mathbb{F}$) have a common eigenbasis. Commuting operators on a complex space can also be made upper-triangular in one common basis (Problem 8).

Why. If $Tv = \lambda v$, then $T(Sv) = S(Tv) = \lambda Sv$, so $Sv \in E(\lambda, T)$. In the same way, S commutes with every $p(T)$: if $p(T)v = 0$ then $p(T)Sv = Sp(T)v = 0$, and $S\,p(T)w = p(T)\,Sw \in \operatorname{range} p(T)$. Common eigenvector: pick an eigenvalue λ of T. S maps $E(\lambda, T) \neq \{0\}$ into itself, so $S|_{E(\lambda,T)}$ is an operator on a nonzero complex space and has an eigenvector w. Then $Sw = \mu w$, and $Tw = \lambda w$ because $w \in E(\lambda, T)$. Common eigenbasis: $V = \bigoplus_\lambda E(\lambda, T)$. On each piece, S restricted to it is diagonalizable (restrictions of diagonalizable operators are diagonalizable, Section 1.14). An eigenbasis of S inside each $E(\lambda, T)$ consists of eigenvectors of both.

Problems.

(a) $T \in \mathcal{L}(\mathbb{C}^3)$ has three distinct eigenvalues and $ST = TS$. Prove that every eigenvector of T is an eigenvector of S, so S is diagonal in T's eigenbasis.

(b) $S^2 = I$ and $ST = TS$ on a real space. Prove $V = E(1, S) \oplus E(-1, S)$ and that both pieces are T-invariant. What does $\mathcal{M}(T)$ look like in a basis adapted to this decomposition?

(c) Show that $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ commutes only with matrices of the form $\left(\begin{smallmatrix} a & b \\ 0 & a \end{smallmatrix}\right)$.

§2.4 Identifying a quotient space

Theorem (T4)

Let $U \subseteq V$ be a subspace.

1. To identify V/U with a familiar space W: find a linear $T\colon V \to W$ that is surjective with $\operatorname{null} T = U$ exactly. Then $v + U \mapsto Tv$ is an isomorphism $V/U \to W$.
2. For a linear $F\colon V \to W$, the rule $v + U \mapsto F(v)$ is a well-defined map on V/U iff $F(u) = 0$ for every $u \in U$.
3. If T is as in (1), then $v_1 + U, \ldots, v_k + U$ is a basis of V/U iff $Tv_1, \ldots, Tv_k$ is a basis of W. Also: extending a basis of U by $v_1, \ldots, v_k$ to a basis of V gives the basis $v_1 + U, \ldots, v_k + U$ of V/U.

Why. (1) is the first isomorphism theorem (Prop. 41). (2) The rule is well defined iff $F(v) = F(v')$ whenever $v - v' \in U$, which for linear F says $F(u) = 0$ for $u \in U$ (Prop. 42 with $U_2 = \{0\}$). (3) An isomorphism sends bases to bases; the second sentence is the basis-of-a-quotient proposition in Section 1.3.

When an exam asks you to "describe the isomorphism given by the first isomorphism theorem", write the formula $v + \operatorname{null} T \mapsto Tv$, say what $\operatorname{null} T$ is, and check surjectivity.

Problems.

(a) Let $V = \mathbb{R}^{2,2}$ and $U = \{A : A^T = A\}$. Prove $V/U \cong \mathbb{R}$, give the isomorphism, and find a basis of V/U.

(b) Let $U = \operatorname{span}\big((1,1,1)\big) \subseteq \mathbb{R}^3$. Give an explicit isomorphism $\mathbb{R}^3/U \to \mathbb{R}^2$ and a basis of $\mathbb{R}^3/U$.

(c) With the same U, which of these rules define maps $\mathbb{R}^3/U \to \mathbb{R}$: (i) $(x, y, z)+U \mapsto x$; (ii) $(x, y, z)+U \mapsto x+y-2z$; (iii) $(x, y, z) + U \mapsto (x - y)^2$?

(d) $T\colon V \to W$ is linear and $U \subseteq \operatorname{null} T$. Prove $\dim \operatorname{range} T \leq \dim V - \dim U$, with equality iff $U = \operatorname{null} T$.

§2.5 Splitting into an invariant subspace and the quotient

Theorem (T5)

Let U be T-invariant, with V finite-dimensional. Then:

1. $m_{T|_U}$ and $m_{T/U}$ both divide m_T, and m_T divides the product $m_{T|_U}\, m_{T/U}$.
2. The eigenvalues of T are exactly the eigenvalues of $T|_U$ together with the eigenvalues of T/U.
3. In a basis $u_1, \ldots, u_k, v_1, \ldots, v_l$ (a basis of U, then extended to V), $\mathcal{M}(T) = \left(\begin{smallmatrix} \mathcal{M}(T|_U) & * \\ 0 & \mathcal{M}(T/U) \end{smallmatrix}\right)$, where $\mathcal{M}(T/U)$ uses the basis $v_1 + U, \ldots, v_l + U$.

Why. (1) For any polynomial p: $p(T|_U)$ is $p(T)$ used on U, and $p(T/U)(v + U) = p(T)v + U$. So $m_T(T) = 0$ gives $m_T(T|_U) = 0$ and $m_T(T/U) = 0$, and Prop. 65 says both minimal polynomials divide m_T. For the

product: $m_{T/U}(T/U) = 0$ means $m_{T/U}(T)v \in U$ for every v, and then $m_{T|_U}(T)$ sends that vector of U to 0. So $m_{T|_U}(T)\, m_{T/U}(T) = 0$. (2) Let $Tv = \lambda v$ with $v \neq 0$. If $v \in U$, λ is an eigenvalue of $T|_U$. If $v \notin U$, then $v+U \neq 0+U$ and $(T/U)(v+U) = \lambda(v+U)$, so λ is an eigenvalue of T/U. Conversely, an eigenvalue of $T|_U$ is clearly one of T. Suppose λ is an eigenvalue of T/U but not of T. Then $T - \lambda I$ is invertible on V; it maps U into U injectively, so onto U (same finite dimension). Now if $(T-\lambda I)v \in U$, write $(T-\lambda I)v = (T-\lambda I)u$ with $u \in U$; then $v = u \in U$. So $(T/U) - \lambda I$, which sends $v+U$ to $(T-\lambda I)v + U$, is injective on V/U, contradicting that λ is an eigenvalue of T/U. (3) $Tu_i \in U$, so the first k columns have zeros below row k. The coefficients of Tv_j on $v_1, \dots, v_l$ are the coefficients of $(T/U)(v_j+U)$ on $v_1+U, \dots, v_l+U$.

Proofs by induction on $\dim V$ use this: apply the induction hypothesis to $T|_U$ and T/U, which act on smaller spaces.

Problems.

(a) (Ribet, Fall 2008) T is invertible and U is T-invariant. If $Tv \in U$, prove $v \in U$.
(b) Give an operator T and invariant U such that $T|_U$ and T/U are both diagonalizable but T is not. Why does this not contradict (1)?
(c) (Ribet, Spring 2020) U is T-invariant and $Tv_j = \lambda_j v_j$ for distinct $\lambda_1, \dots, \lambda_k$. If $\sum_j (v_j + U) = 0 + U$, prove each $v_j \in U$.

§2.6 Cyclic subspaces

Theorem (T6)

For $v \neq 0$, let k be the largest number such that $v, Tv, \dots, T^{k-1}v$ is linearly independent, and let $W = \operatorname{span}(v, \dots, T^{k-1}v)$. Then W is T-invariant, it is the smallest T-invariant subspace containing v, and the minimal polynomial of $T|_W$ has degree exactly k. If some v has $W = V$, v is called a **cyclic vector**; then $\deg m_T = \dim V$.

Why. By the choice of k, $T^k v \in W$, so T maps each of $v, \dots, T^{k-1}v$ into W; hence W is invariant. Any invariant subspace containing v contains Tv, T^2v, and so on, so it contains W. If $q(T|_W) = 0$ with $\deg q < k$, then $q(T)v = 0$ is a dependence among $v, \dots, T^{k-1}v$, which is impossible; so $\deg m_{T|_W} \geq k$. And $\deg m_{T|_W} \leq \dim W = k$ by Prop. 67.

Problems.

(a) (Ribet, Fall 2003) $T^m v = 0$ but $T^{m-1}v \neq 0$. Prove $v, Tv, \dots, T^{m-1}v$ is independent.
(b) T is diagonalizable and has a cyclic vector. Prove that T has $\dim V$ distinct eigenvalues.
(c) Find a cyclic vector for $\operatorname{diag}(1,2,3)$, and prove that I_2 on $\mathbb{F}^2$ has none.

§2.7 Products, sums and graphs

Theorem (T7)

1. $\dim(V_1 \times \cdots \times V_m) = \dim V_1 + \cdots + \dim V_m$; a basis is obtained by putting a basis of each V_i in its own slot, with zeros in the other slots.
2. For subspaces $U_1, \dots, U_m$, $\Gamma(u_1, \dots, u_m) = u_1 + \cdots + u_m$ maps $U_1 \times \cdots \times U_m$ onto $U_1 + \cdots + U_m$, and it is injective iff the sum is direct. So the sum is direct iff $\dim(U_1 + \cdots + U_m) = \dim U_1 + \cdots + \dim U_m$.
3. For two subspaces, $\operatorname{null}\Gamma = \{(u, -u) : u \in U \cap W\}$, which is isomorphic to $U \cap W$.
4. A map into a product, $v \mapsto (T_1 v, \dots, T_m v)$, is linear iff each T_i is linear.

Why. (1)–(3) are in Section 1.2. (4) Addition and scaling in a product are done entry by entry.

Problems.

(a) In $\mathcal{P}_2(\mathbb{R})$ let $U = \{p : p(1) = 0\}$ and $W = \{p : p(-1) = 0\}$. Find a basis of $U \cap W$, and use Γ to find $\dim(U+W)$.
(b) Prove: in finite dimensions, $U_1 + \cdots + U_m$ is a direct sum iff $\dim(U_1 + \cdots + U_m) = \dim U_1 + \cdots + \dim U_m$.
(c) $T \in \mathcal{L}(V, W)$. Prove that the graph $G = \{(v, Tv) : v \in V\}$ is a subspace of $V \times W$, that $\dim G = \dim V$, and that $(V \times W)/G \cong W$.
(d) Prove $\mathcal{L}(V_1 \times V_2, W) \cong \mathcal{L}(V_1, W) \times \mathcal{L}(V_2, W)$.

§2.8 Duality facts

Theorem (T8)

In finite dimensions:

1. $\dim U^0 = \dim V - \dim U$; $U \subseteq W \iff W^0 \subseteq U^0$; $U^0 = \{0\} \iff U = V$.
2. $(U+W)^0 = U^0 \cap W^0$ and $(U \cap W)^0 = U^0 + W^0$.
3. $\operatorname{null} T' = (\operatorname{range} T)^0$, $\operatorname{range} T' = (\operatorname{null} T)^0$, $\operatorname{rank} T' = \operatorname{rank} T$.
4. U is T-invariant iff U^0 is T'-invariant.
5. T and T' have the same eigenvalues with the same geometric multiplicities.

Why. (1), (2) and (3) are proved in Section 1.9; (5) is in Section 1.11 (eigenvalues of the dual map). Problem (a) asks you to reproduce the proof of the second half of (2). (4): if U is invariant and $\varphi \in U^0$, then for $u \in U$, $(T'\varphi)(u) = \varphi(Tu) = 0$ because $Tu \in U$; so $T'\varphi \in U^0$. The converse is problem (b).

Problems.

(a) Prove $(U \cap W)^0 = U^0 + W^0$ in full.
(b) Prove (4), including the converse.
(c) (Ribet, Spring 2014) $U \subsetneq \mathbb{F}^n$ is a proper subspace. Prove there are $c_1, \dots, c_n$, not all 0, with $\sum c_i a_i = 0$ for every $(a_1, \dots, a_n) \in U$.
(d) $\varphi_1, \dots, \varphi_m \in V'$. Prove $\dim \bigcap_i \operatorname{null} \varphi_i \geq \dim V - m$, with equality iff the φ_i are independent.

§2.9 Computing powers

Theorem (T9)

If $A = PDP^{-1}$ with D diagonal, then $A^k = PD^kP^{-1}$. If $A = \lambda I + N$ with $N^r = 0$, then $A^k = \sum_{j=0}^{\min(k,r-1)} \binom{k}{j} \lambda^{k-j} N^j$. In particular, if $N^2 = 0$: $A^k = \lambda^k I + k\lambda^{k-1} N$.

Why. In $(PDP^{-1})(PDP^{-1}) \cdots (PDP^{-1})$ each inner $P^{-1}P$ cancels. The binomial theorem works for $(\lambda I + N)^k$ because λI and N commute, and the terms with N^j, $j \geq r$, are 0.

Problems.

(a) $A = \left(\begin{smallmatrix} 3 & 1 \\ -1 & 1 \end{smallmatrix}\right)$. Find A^k. (A is not diagonalizable.)
(b) Using $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, derive $F_n = \frac{g_+^n - g_-^n}{\sqrt{5}}$ for the Fibonacci numbers ($F_0 = 0$, $F_1 = 1$), where $g_\pm = \frac{1 \pm \sqrt{5}}{2}$.
(c) For which λ does $\left(\begin{smallmatrix} \lambda & 1 \\ 0 & \lambda \end{smallmatrix}\right)^k \to 0$ as $k \to \infty$? Compare with $\operatorname{diag}(\lambda, \lambda)$.

§2.10 Standard counterexamples

T10. Most false statements in this material are refuted by one of these:

Rotation $R = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$	no real eigenvalue; $m_R = x^2 + 1$; diagonalizable over $\mathbb{C}$, not over $\mathbb{R}$; $R^4 = I$.
$N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$	only eigenvalue 0 but $N \neq 0$; not diagonalizable; $\operatorname{null} N = \operatorname{range} N$, so $V \neq \operatorname{null} N \oplus \operatorname{range} N$; same χ as the zero matrix; $N^2 = 0$ is diagonalizable but N is not.
$\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ versus I_2	same eigenvalues and χ, not similar; the first is invertible but not diagonalizable.
$p \mapsto xp$ on $\mathcal{P}(\mathbb{C})$	no eigenvalue: existence of eigenvalues needs finite dimension.
Backward shift $(a_1, a_2, a_3, \dots) \mapsto (a_2, a_3, \dots)$ on infinite sequences	every scalar λ is an eigenvalue, with eigenvector $(1, \lambda, \lambda^2, \dots)$: "at most $\dim V$ eigenvalues" needs finite dimension.
$\operatorname{diag}(1, -1)$	$T^2 = I$ with $T \neq \pm I$.
$v + U$ with $v \notin U$	an affine subset that is not a subspace (it does not contain 0).
$V = \{0\}$	$T^3 = T$ but no eigenvalue at all: watch for "$V \neq \{0\}$".

Problems.

(a) True or false, naming a counterexample if false: (i) if 0 is the only eigenvalue of T, then $T = 0$; (ii) if T^2 is diagonalizable then T is; (iii) if U is T-invariant and $T|_U$ and T/U are diagonalizable, then T is; (iv) every operator on a real space has an invariant subspace of dimension 1; (v) every affine subset of V is a subspace.

§2.11 Facts you may quote

- Eigenvectors for distinct eigenvalues are independent; there are at most $\dim V$ eigenvalues.
- V/U: $v+U = w+U \iff v-w \in U$; $\dim V/U = \dim V - \dim U$; first isomorphism theorem $V/\operatorname{null} T \cong \operatorname{range} T$; second and third isomorphism theorems; $\dim(V_1 \times \cdots \times V_m) = \sum \dim V_i$.
- Over $\mathbb{C}$, every operator on a nonzero finite-dimensional space has an eigenvalue and an upper-triangular matrix.
- Upper-triangular matrix: the eigenvalues are the diagonal entries.
- T diagonalizable $\iff \sum_\lambda \dim E(\lambda, T) = \dim V \iff m_T$ splits with no repeated root. $\dim V$ distinct eigenvalues $\Rightarrow$ diagonalizable.
- $q(T) = 0 \iff m_T$ divides q; the roots of m_T that lie in $\mathbb{F}$ are exactly the eigenvalues; $\deg m_T \le \dim V$.
- $\dim U^0 = \dim V - \dim U$; $\operatorname{null} T' = (\operatorname{range} T)^0$; $\operatorname{range} T' = (\operatorname{null} T)^0$; $\mathcal{M}(T') = \mathcal{M}(T)^T$.

§3 Drills

§3.1 Products, affine subsets and quotient spaces (Ch. 8)

1. A product space. Let $V = \mathcal{P}_1(\mathbb{R}) \times \mathbb{R}^2$.

a) (2) Give a basis of V and its dimension.
b) (3) Define $T\colon V \to \mathcal{P}_3(\mathbb{R})$ by $T\big(p,(a,b)\big) = p + ax^2 + bx^3$. Prove that T is an isomorphism.
c) (3) Find the dimension of $\{(p,(a,b)) \in V : p(1) = a + b\}$.

2. The sum map. Let $U_1, \dots, U_m$ be subspaces of V and $\Gamma\colon U_1 \times \cdots \times U_m \to V$, $\Gamma(u_1, \dots, u_m) = u_1 + \cdots + u_m$.

a) (2) Prove that Γ is linear.
b) (2) Prove that $\operatorname{range}\Gamma = U_1 + \cdots + U_m$.
c) (3) Prove that Γ is injective if and only if the sum $U_1 + \cdots + U_m$ is direct.
d) (3) Assume V is finite-dimensional. Deduce $\dim(U_1 + \cdots + U_m) \le \dim U_1 + \cdots + \dim U_m$, with equality if and only if the sum is direct.

3. Affine subsets. Let U be a subspace of V and $v, w \in V$.

a) (3) Prove that if $v - w \in U$, then $v + U = w + U$.
b) (3) Prove that if $(v + U) \cap (w + U) \neq \varnothing$, then $v - w \in U$.
c) (2) Prove that if $v + U = x + W$ for a vector x and a subspace W, then $U = W$.
d) (2) Let $A = \{(x,y,z) \in \mathbb{R}^3 : x + 2y - z = 3\}$. Find the subspace U to which A is parallel and a vector v with $A = v + U$.

4. Solution sets. Let $T\colon \mathbb{R}^3 \to \mathbb{R}^2$, $T(x,y,z) = (x + y + z,\ x - z)$.

a) (2) Find $\operatorname{null} T$.
b) (3) Write $\{u \in \mathbb{R}^3 : Tu = (2,1)\}$ in the form $v + \operatorname{null} T$.
c) (3) Prove in general: for $T \in \mathcal{L}(V,W)$ and $b \in W$, the set $\{u : Tu = b\}$ is either empty or equal to $u_0 + \operatorname{null} T$ for any u_0 with $Tu_0 = b$.

5. Operations on V/U. Let U be a subspace of V.

a) (3) Prove that $(v + U) + (w + U) = (v + w) + U$ is well defined: if $v + U = v' + U$ and $w + U = w' + U$, the two answers agree.
b) (2) Prove that $\lambda(v + U) = \lambda v + U$ is well defined.
c) (3) Let W be a subspace with $U \subseteq W \subseteq V$. Prove that $W/U = \{w + U : w \in W\}$ is a subspace of V/U, and if V is finite-dimensional, that $\dim W/U = \dim W - \dim U$.
d) (2) In $\mathbb{R}^3/U$ with $U = \operatorname{span}(e_3)$, decide whether $(1,2,5) + U = (1,2,-1) + U$ and whether $(1,2,5) + U = (1,3,5) + U$.

6. A basis of a quotient. Let $V = \mathbb{R}^4$ and $U = \operatorname{span}\big((1,1,0,0),(0,0,1,1)\big)$.

a) (3) Find a basis of V/U.
b) (3) Write $(3,1,4,1) + U$ in your basis.
c) (2) Find a linear $T\colon \mathbb{R}^4 \to \mathbb{R}^2$ with $\operatorname{null} T = U$, and use it to write down an isomorphism $V/U \to \mathbb{R}^2$.

7. Quotients of an infinite-dimensional space. Let $V = \mathcal{P}(\mathbb{R})$.

a) (3) Let $U = \{p : p(0) = 0\}$. Prove V/U is 1-dimensional and give a basis.
b) (3) Let $W = \{p : p(0) = p'(0) = 0\}$. Find $\dim V/W$.
c) (2) Prove $V/\{0\} \cong V$.

§3.2 Quotient maps and the isomorphism theorems (Ch. 9)

8. The first isomorphism theorem. Let $T \in \mathcal{L}(V, W)$.

a) (2) Prove that $\bar{T}(v + \operatorname{null} T) = Tv$ is well defined on $V/\operatorname{null} T$.
b) (2) Prove that $\bar{T}$ is linear.
c) (3) Prove that $\bar{T}$ is injective.
d) (3) Prove that $\bar{T}$ maps onto $\operatorname{range} T$. Apply this to differentiation on $\mathcal{P}(\mathbb{R})$ to show $\mathcal{P}(\mathbb{R})/\mathbb{R} \cong \mathcal{P}(\mathbb{R})$, where $\mathbb{R}$ is the subspace of constant polynomials.

9. Symmetric parts. Let $V = \mathbb{R}^{2,2}$ and $T(A) = A + A^T$.

a) (3) Find $\operatorname{null} T$ and $\operatorname{range} T$.
b) (3) Write down the isomorphism $V/\operatorname{null} T \to \operatorname{range} T$ given by the first isomorphism theorem.
c) (2) Find a basis of $V/\operatorname{null} T$ and its image.
d) (2) Check that the dimensions agree.

10. Taylor coefficients. Let $V = \mathcal{P}_4(\mathbb{R})$, $T(p) = \big(p(2), p'(2), p''(2)\big)$, and $U = \operatorname{null} T$.

a) (3) Prove $T\colon V \to \mathbb{R}^3$ is surjective and $U = \{(x-2)^3 q : q \in \mathcal{P}_1(\mathbb{R})\}$.
b) (2) Write down the isomorphism $V/U \to \mathbb{R}^3$.
c) (3) Prove that $1 + U,\ (x-2) + U,\ (x-2)^2 + U$ is a basis of V/U, and find their images.
d) (2) Write $x^4 + U$ in this basis.

11. Maps out of a quotient. Let $T \in \mathcal{L}(V, W)$ and let U be a subspace with $U \subseteq \operatorname{null} T$.

a) (3) Prove there is exactly one linear $\tilde{T}\colon V/U \to W$ with $\tilde{T}(v + U) = Tv$ for all v.
b) (3) Prove $\operatorname{null} \tilde{T} = \{v + U : v \in \operatorname{null} T\}$.
c) (2) Prove $\tilde{T}$ is injective if and only if $U = \operatorname{null} T$.
d) (2) For $T(x, y, z) = x - y$ on $\mathbb{R}^3$ and $U = \operatorname{span}(e_3)$, find $\operatorname{null} \tilde{T}$. Is $\tilde{T}$ injective?

12. The second isomorphism theorem. Let U, W be subspaces of V.

a) (3) Prove that $S\colon U \to (U+W)/W$, $S(u) = u + W$, is linear and surjective.
b) (3) Prove $\operatorname{null} S = U \cap W$.
c) (2) Deduce $U/(U \cap W) \cong (U+W)/W$, and, for V finite-dimensional, $\dim(U+W) = \dim U + \dim W - \dim(U \cap W)$.
d) (2) For $U = \operatorname{span}(e_1, e_2)$ and $W = \operatorname{span}(e_2 + e_3)$ in $\mathbb{R}^3$, find $U \cap W$ and $U + W$ and check both sides.

13. The third isomorphism theorem. Let $W \subseteq U \subseteq V$ be subspaces.

a) (2) Prove that U/W is a subspace of V/W.
b) (3) Prove that $S\colon V/W \to V/U$, $S(v + W) = v + U$, is well defined, linear and surjective.
c) (3) Prove $\operatorname{null} S = U/W$ and deduce $(V/W)/(U/W) \cong V/U$.
d) (2) For $V = \mathbb{R}^3$, $W = \operatorname{span}(e_1)$, $U = \operatorname{span}(e_1, e_2)$, identify $(V/W)/(U/W)$ with $\mathbb{R}$ by an explicit formula.

14. A quotient operator. Let $T \in \mathcal{L}(\mathbb{R}^3)$ have matrix $A = \left(\begin{smallmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 1 & 1 & 3 \end{smallmatrix}\right)$ in the standard basis, and $U = \operatorname{span}(e_3)$.

a) (2) Prove U is T-invariant.
b) (3) Find the matrix of T/U in the basis $e_1 + U,\ e_2 + U$.
c) (3) Find the eigenvalues of $T|_U$, T/U and T. Is T/U diagonalizable?
d) (2) Without computing eigenvectors of T, decide whether T is diagonalizable.

§3.3 Duality (Ch. 10)

15. Dual bases. Let $v_1, \ldots, v_n$ be a basis of V with dual basis $\varphi_1, \ldots, \varphi_n$.

a) (3) Prove $\varphi_1, \ldots, \varphi_n$ is linearly independent.
b) (2) Deduce that it is a basis of V'.
c) (3) Prove $\psi = \sum_j \psi(v_j)\varphi_j$ for every $\psi \in V'$.
d) (2) Find the dual basis of $(1, 2), (3, 4)$ in $\mathbb{R}^2$.

16. Functionals on $\mathcal{P}_2(\mathbb{R})$. Use the basis $1,\ x-1,\ (x-1)^2$ of $V=\mathcal{P}_2(\mathbb{R})$.

a) (3) Prove that the dual basis is $\varphi_1(p)=p(1)$, $\varphi_2(p)=p'(1)$, $\varphi_3(p)=\frac{1}{2}p''(1)$.
b) (3) Write $\psi(p)=\int_0^1 p(x)\,dx$ in this dual basis.
c) (2) Write $\varepsilon(p)=p(0)$ in this dual basis.
d) (2) Is $\varphi_1,\psi,\varepsilon$ a basis of V'?

17. The double dual. For $v\in V$ define $j_V(v)\colon V'\to\mathbb{F}$ by $j_V(v)(\varphi)=\varphi(v)$.

a) (2) Prove $j_V(v)$ is linear, so $j_V(v)\in V''$.
b) (2) Prove $j_V\colon V\to V''$ is linear.
c) (3) Prove j_V is injective.
d) (1) Deduce that j_V is an isomorphism when V is finite-dimensional.

18. The dual of differentiation. Let $T\colon\mathcal{P}_2(\mathbb{R})\to\mathcal{P}_1(\mathbb{R})$, $Tp=p'$. Let $\varphi_0,\varphi_1,\varphi_2$ be the dual basis of $1,x,x^2$ and ψ_0,ψ_1 the dual basis of $1,x$.

a) (3) Find $\mathcal{M}(T)$ and $\mathcal{M}(T')$.
b) (3) Compute $T'(\psi_0)$ and $T'(\psi_1)$ directly from the definition of the dual map.
c) (2) Find $\operatorname{null}T'$ and $\operatorname{range}T'$, and check them against $\operatorname{null}T'=(\operatorname{range}T)^0$ and $\operatorname{range}T'=(\operatorname{null}T)^0$.
d) (2) Let $\varepsilon(q)=q(1)$ on $\mathcal{P}_1(\mathbb{R})$. Write $T'(\varepsilon)$ in terms of the φ_j.

19. Properties of the dual map. Let S,T be linear maps.

a) (2) Prove $(S+T)'=S'+T'$ and $(cT)'=cT'$ for $S,T\in\mathcal{L}(V,W)$.
b) (3) Prove $(ST)'=T'S'$ for $T\in\mathcal{L}(U,V)$, $S\in\mathcal{L}(V,W)$.
c) (3) Prove that if $T\in\mathcal{L}(V,W)$ is invertible, then T' is invertible and $(T')^{-1}=(T^{-1})'$.
d) (2) Find $\mathcal{M}(T')$ in the dual bases for $T(x,y)=(x+y,\ 2x,\ 3y)$, $T\colon\mathbb{R}^2\to\mathbb{R}^3$.

20. Computing annihilators.

a) (3) Find a basis of U^0 for $U=\operatorname{span}\big((1,0,1,0),(0,1,0,1)\big)\subseteq\mathbb{R}^4$ (functionals as row vectors).
b) (3) In $\mathcal{P}_3(\mathbb{R})$ let $U=\{p:p(1)=p(2)=0\}$ and $\varepsilon_a(p)=p(a)$. Prove $U^0=\operatorname{span}(\varepsilon_1,\varepsilon_2)$.
c) (2) Let $W=\mathcal{P}_1(\mathbb{R})\subseteq\mathcal{P}_3(\mathbb{R})$. Is $\varepsilon_1-\varepsilon_2\in W^0$?
d) (2) Find $\dim W^0$ and one nonzero element of W^0.

21. Null space and range of T'. Let $T\in\mathcal{L}(V,W)$ with V,W finite-dimensional.

a) (3) Prove $\operatorname{null}T'=(\operatorname{range}T)^0$.
b) (2) Prove $\operatorname{range}T'\subseteq(\operatorname{null}T)^0$.
c) (3) Prove $\operatorname{range}T'=(\operatorname{null}T)^0$ by comparing dimensions.
d) (2) Deduce that T is surjective iff T' is injective, and that a matrix and its transpose have the same rank.

22. Annihilators in general. Let V be finite-dimensional and U,W subspaces.

a) (3) Prove that the restriction map $V'\to U'$, $\varphi\mapsto\varphi|_U$, is surjective with null space U^0, and deduce $\dim U^0=\dim V-\dim U$.
b) (3) Prove $U\subseteq W$ if and only if $W^0\subseteq U^0$.
c) (2) Prove $(U+W)^0=U^0\cap W^0$.
d) (2) Prove: if $\varphi_1,\dots,\varphi_m\in V'$ and $\operatorname{null}\varphi_1\cap\cdots\cap\operatorname{null}\varphi_m=\{0\}$, then $m\ge\dim V$.

23. When functionals form a basis. Let $\dim V=n$ and $\varphi_1,\dots,\varphi_n\in V'$; define $S\colon V\to\mathbb{F}^n$, $S(v)=\big(\varphi_1(v),\dots,\varphi_n(v)\big)$.

a) (3) Prove that $\varphi_1,\dots,\varphi_n$ is linearly independent iff S is surjective. (Hint: S' sends the ith coordinate functional on $\mathbb{F}^n$ to φ_i.)
b) (3) Deduce that $\varphi_1,\dots,\varphi_n$ is a basis of V' iff the only v with $\varphi_i(v)=0$ for all i is $v=0$.
c) (2) Decide whether $p\mapsto p(0)$, $p\mapsto p(1)$, $p\mapsto p(2)$ is a basis of $\mathcal{P}_2(\mathbb{R})'$.

§3.4 Invariant subspaces and eigenvalues (Ch. 11)

24. Invariant subspaces of a 3×3 matrix. Let $T(x,y,z)=(2x,\ x+2y,\ 5z)$ on $\mathbb{R}^3$.

a) (3) Find the eigenvalues and eigenspaces of T.
b) (3) Decide which of $\operatorname{span}(e_1)$, $\operatorname{span}(e_1,e_2)$, $\operatorname{span}(e_2,e_3)$ are T-invariant.
c) (2) List all 1-dimensional invariant subspaces.
d) (2) Is T diagonalizable?

25. Existence of eigenvalues. Let $V\neq\{0\}$ be a finite-dimensional complex vector space, $n=\dim V$, and $T\in\mathcal{L}(V)$.

a) (2) Fix $v\neq 0$. Prove there is a nonconstant polynomial p of degree at most n with $p(T)v=0$.
b) (3) Choose such p of smallest degree and factor $p=(x-\lambda)q$. Explain why this factorization exists and why $q(T)v\neq 0$.
c) (3) Prove that $w=q(T)v$ is an eigenvector of T.
d) (2) Where does this argument break down for the rotation $R(x,y)=(-y,x)$ on $\mathbb{R}^2$?

26. Independence of eigenvectors. Let $T\in\mathcal{L}(V)$.

a) (4) Prove that eigenvectors $v_1,\dots,v_m$ for distinct eigenvalues $\lambda_1,\dots,\lambda_m$ are linearly independent.
b) (2) Deduce that T has at most $\dim V$ distinct eigenvalues if V is finite-dimensional.
c) (2) Prove that e^x, e^{2x}, e^{3x} are linearly independent functions.
d) (2) Show that every $\lambda\in\mathbb{R}$ is an eigenvalue of the backward shift $(a_1,a_2,a_3,\dots)\mapsto(a_2,a_3,\dots)$ on real sequences. Why does this not contradict (b)?

27. A reflection of polynomials. Let $T\colon \mathcal{P}_3(\mathbb{R})\to\mathcal{P}_3(\mathbb{R})$, $(Tp)(x)=p(-x)$.

a) (2) Prove $T^2=I$ and deduce that every eigenvalue is 1 or -1.
b) (3) Find $E(1,T)$ and $E(-1,T)$.
c) (3) Prove T is diagonalizable in two ways: by dimensions of eigenspaces, and by the minimal polynomial.
d) (2) Find the eigenvalues of T' and the dimensions of its eigenspaces.

28. Diagonalizing a 2×2 matrix. Let $A=\left(\begin{smallmatrix}4 & -2\\ 1 & 1\end{smallmatrix}\right)$.

a) (3) Find the eigenvalues and eigenvectors of A.
b) (3) Find P invertible and D diagonal with $A=PDP^{-1}$.
c) (2) Find a formula for A^k.
d) (2) Find m_A.

29. Eigenvalues that survive changes.

a) (3) Let S be invertible and $R=S^{-1}TS$. Prove T and R have the same eigenvalues and $E(\lambda,R)=S^{-1}\big(E(\lambda,T)\big)$.
b) (3) Prove that T and T' have the same eigenvalues with the same geometric multiplicities (V finite-dimensional).
c) (2) For $A=\left(\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right)$ find $E(1,A)$ and $E(1,A^T)$. Are they equal?

30. Stochastic matrices. Let P be an $n\times n$ column-stochastic matrix.

a) (3) Prove every eigenvalue λ of P satisfies $|\lambda|\le 1$.
b) (3) Prove that 1 is an eigenvalue of P. (Hint: $P^T\mathbf{1}=\mathbf{1}$ for $\mathbf{1}=(1,\dots,1)$.)
c) (2) Find the stationary distribution of $P=\left(\begin{smallmatrix}1/2 & 1/5\\ 1/2 & 4/5\end{smallmatrix}\right)$.
d) (2) Find the other eigenvalue of this P.

31. PageRank. Let P be column-stochastic, v a probability vector, $0<\alpha<1$.

a) (3) Prove $I-\alpha P$ is invertible.
b) (3) For $P=\left(\begin{smallmatrix}0 & 1\\ 1 & 0\end{smallmatrix}\right)$, $v=(1,0)$, $\alpha=\frac{1}{2}$, solve $r=\alpha Pr+(1-\alpha)v$.
c) (2) Check that your r is a probability vector and explain in words why $r_1>r_2$.

32. Eigenvalues of restriction and quotient. Let V be finite-dimensional, $T\in\mathcal{L}(V)$, U a T-invariant subspace.

a) (3) Prove every eigenvalue of $T|_U$ is an eigenvalue of T.
b) (3) Prove: if $Tv=\lambda v$ with $v\notin U$, then λ is an eigenvalue of T/U.
c) (4) Suppose λ is not an eigenvalue of T. Prove $T-\lambda I$ maps U onto U, and deduce that $(T/U)-\lambda I$ is injective. Conclude that every eigenvalue of T/U is an eigenvalue of T.

§3.5 Upper-triangular matrices and diagonalizability (Ch. 12)

33. Upper-triangular matrices. Let $v_1, \dots, v_n$ be a basis of V and $T \in \mathcal{L}(V)$.

a) (3) Prove: if $\mathcal{M}(T)$ is upper-triangular, then $Tv_j \in \operatorname{span}(v_1, \dots, v_j)$ for each j.
b) (3) Prove: if $Tv_j \in \operatorname{span}(v_1, \dots, v_j)$ for each j, then each $\operatorname{span}(v_1, \dots, v_j)$ is T-invariant.
c) (2) Prove: if each $\operatorname{span}(v_1, \dots, v_j)$ is T-invariant, then $\mathcal{M}(T)$ is upper-triangular.

34. Eigenvalues on the diagonal. Let $\mathcal{M}(T)$ be upper-triangular with diagonal $\lambda_1, \dots, \lambda_n$.

a) (4) Prove that λ is an eigenvalue of T iff $\lambda = \lambda_k$ for some k.
b) (2) Deduce that T is invertible iff no λ_k is 0.
c) (2) Let $T = I + D$ on $\mathcal{P}_2(\mathbb{R})$, with D differentiation. Find the eigenvalues and eigenspaces of T.
d) (2) Write T^{-1} as a polynomial in D.

35. Triangularizing a matrix. Let $A = \left(\begin{smallmatrix} 3 & 1 \\ -4 & -1 \end{smallmatrix}\right)$.

a) (3) Find the eigenvalues and eigenvectors of A.
b) (4) Find a basis in which the operator $x \mapsto Ax$ has an upper-triangular matrix, and the matrix S with $S^{-1}AS$ upper-triangular.
c) (3) Prove A is not diagonalizable and find m_A.

36. A diagonalizability test with a parameter. For $t \in \mathbb{R}$ let $A_t = \left(\begin{smallmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ -3 & t & 2 \end{smallmatrix}\right)$.

a) (2) Find the eigenvalues of A_t.
b) (4) Find all t for which A_t is diagonalizable.
c) (4) For those t, find an eigenbasis.

37. Conditions for diagonalizability. Let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues of $T \in \mathcal{L}(V)$, V finite-dimensional.

a) (2) Prove: $\mathcal{M}(T)$ is diagonal in a basis iff that basis consists of eigenvectors.
b) (3) Prove the sum $E(\lambda_1, T) + \cdots + E(\lambda_m, T)$ is direct.
c) (3) Prove V has an eigenbasis iff $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$.
d) (2) Prove this holds iff $\dim V = \sum_i \dim E(\lambda_i, T)$.

38. Projections. Let $V \neq \{0\}$ be finite-dimensional and $P \in \mathcal{L}(V)$ with $P^2 = P$.

a) (3) Prove $V = \operatorname{null} P \oplus \operatorname{range} P$.
b) (2) Prove $\operatorname{range} P = E(1, P)$.
c) (3) Prove P is diagonalizable, and list the possible minimal polynomials of P.
d) (2) Find the matrix of the projection of $\mathbb{R}^2$ onto $\operatorname{span}\big((1,1)\big)$ along $\operatorname{span}\big((1,0)\big)$.

39. A recurrence. Let $a_0 = 0$, $a_1 = 1$, $a_{n+2} = a_{n+1} + 2a_n$.

a) (3) Find a 2×2 matrix A with $A\left(\begin{smallmatrix} a_n \\ a_{n+1} \end{smallmatrix}\right) = \left(\begin{smallmatrix} a_{n+1} \\ a_{n+2} \end{smallmatrix}\right)$, and its eigenvalues and eigenvectors.
b) (4) Write $\left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right)$ in the eigenbasis.
c) (3) Deduce a formula for a_n, and check it for $n = 2, 3$.

40. A two-state Markov chain. Let $P = \left(\begin{smallmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{smallmatrix}\right)$.

a) (3) Find the eigenvalues and eigenvectors of P.
b) (3) Write $p_0 = (1, 0)$ in the eigenbasis, with the eigenvector for 1 scaled to be a probability vector.
c) (2) Find $P^k p_0$.
d) (2) Find $\lim_k P^k p_0$, and say how fast the error shrinks.

§3.6 The minimal polynomial (Ch. 13)

41. Minimal polynomials from the definition. Let $A = \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$ and $B = \operatorname{diag}(1, 1, 2)$.

a) (3) Find m_A directly from the definition (the first power of A that is a combination of lower powers).
b) (3) Find m_B.
c) (2) Which of A, B is diagonalizable?
d) (2) Find m_{A-I} and m_{2B}.

42. Polynomials that annihilate T. Let V be finite-dimensional and $T \in \mathcal{L}(V)$.

a) (4) Prove that $q(T) = 0$ iff m_T divides q.
b) (3) Let V be complex and $T^4 = T$. Prove T is diagonalizable and list its possible eigenvalues.
c) (3) Give $T \in \mathcal{L}(\mathbb{R}^2)$ with $T^4 = T$ that is not diagonalizable over $\mathbb{R}$. Which real numbers can be eigenvalues of a real T with $T^4 = T$?

43. The degree bound. Let $\dim V = n$ and $T \in \mathcal{L}(V)$. Prove $\deg m_T \le n$ by induction on n.

a) (2) For $v \neq 0$, find a nonzero q of degree at most n with $q(T)v = 0$. Handle the case $q(T) = 0$.
b) (3) If $q(T) \neq 0$, prove $U = \operatorname{null} q(T)$ is T-invariant and $\{0\} \neq U \neq V$.
c) (3) Prove m_T divides $m_{T|_U}\, m_{T/U}$.
d) (2) Finish the induction.

44. Distinct roots give diagonalizability. Let V be finite-dimensional and $T \in \mathcal{L}(V)$ with $m_T = (x-\lambda_1)\cdots(x-\lambda_m)$ with distinct $\lambda_i \in \mathbb{F}$, $m \ge 2$, and $g(x) = (x-\lambda_2)\cdots(x-\lambda_m)$.

a) (3) Show $g = (x-\lambda_1)q + s$ for a polynomial q and a nonzero constant s.
b) (3) For $v \in V$ let $u = \frac{1}{s}g(T)v$. Prove $u \in \operatorname{null}(T-\lambda_1 I)$ and $v - u \in \operatorname{range}(T-\lambda_1 I)$.
c) (2) Deduce $V = \operatorname{null}(T-\lambda_1 I) \oplus \operatorname{range}(T-\lambda_1 I)$.
d) (2) Finish the proof that T is diagonalizable by induction on $\dim V$.

45. A companion matrix. Let $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 6 & -11 & 6 \end{pmatrix}$.

a) (3) Show that e_1, Ae_1, A^2e_1 is a basis, and find m_A.
b) (2) Is A diagonalizable over $\mathbb{R}$?
c) (3) Write A^{-1} as a polynomial in A.
d) (2) Write A^4 as a combination of I, A, A^2.

46. Minimal polynomial of a direct sum. Let V be finite-dimensional, $T \in \mathcal{L}(V)$, and $V = U \oplus W$ with U, W both T-invariant.

a) (3) Prove $p(T) = 0$ iff $p(T|_U) = 0$ and $p(T|_W) = 0$.
b) (3) Deduce that m_T is the monic polynomial of least degree divisible by both $m_{T|_U}$ and $m_{T|_W}$.
c) (2) Deduce that T is diagonalizable iff $T|_U$ and $T|_W$ are.
d) (2) Find m_T for the block matrix $\begin{pmatrix} 1 & 1 & & \\ 0 & 1 & & \\ & & 2 & 0 \\ & & 0 & 1 \end{pmatrix}$.

47. An operator on polynomials. Let $T\colon \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_3(\mathbb{R})$, $(Tp)(x) = xp'(x)$.

a) (3) Find $\mathcal{M}(T)$ in the basis $1, x, x^2, x^3$ and the eigenvalues of T.
b) (3) Find m_T.
c) (2) Find m_{T^2}.
d) (2) Find a polynomial p such that $p(T)$ is the projection onto $\operatorname{span}(x^2)$ along $\operatorname{span}(1, x, x^3)$.

48. Real operators. Let $V \neq \{0\}$ be a finite-dimensional real vector space and $T \in \mathcal{L}(V)$.

a) (4) Prove T has an invariant subspace of dimension 1 or 2. (Factor m_T over $\mathbb{R}$.)
b) (2) Prove that if $\dim V$ is odd, T has an eigenvalue. (Induct, using (a) and the quotient.)
c) (2) Let $T(x, y, z, w) = (-y, x, -w, z)$ on $\mathbb{R}^4$. Show T has no eigenvalue, and give a 2-dimensional invariant subspace.

§4 Practice midterms

§4.1 Practice Midterm A (40 points, 50 minutes)

1. A dual basis and an annihilator. In $\mathbb{R}^3$, identify functionals with row vectors.

a) (3) Let $U = \operatorname{span}\big((1,2,0),(0,1,1)\big)$. Find a basis of U^0.
b) (3) Find the dual basis $\varphi_1, \varphi_2, \varphi_3$ of $v_1 = (1,1,0)$, $v_2 = (0,1,1)$, $v_3 = (1,0,1)$.
c) (2) Write $\psi(x,y,z) = x - 2y + 3z$ as a combination of $\varphi_1, \varphi_2, \varphi_3$.

2. An operator with $T^3 = T$. Let V be a finite-dimensional real vector space and $T \in \mathcal{L}(V)$ with $T^3 = T$.

a) (2) Prove that every eigenvalue of T lies in $\{0, 1, -1\}$.
b) (3) For $v \in V$, put $a = v - T^2 v$, $b = \frac{1}{2}(T^2 v + Tv)$, $c = \frac{1}{2}(T^2 v - Tv)$. Prove $v = a + b + c$ with $Ta = 0$, $Tb = b$, $Tc = -c$.
c) (3) Deduce that T is diagonalizable.
d) (2) Let $T(x,y,z) = (0,z,y)$ on $\mathbb{R}^3$. Verify $T^3 = T$, find the three eigenspaces, and find m_T.

3. Minimal polynomial of a 3×3 matrix. Let

$$A = \begin{pmatrix} 1 & 1 & 0 \\ -1 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

a) (3) Find the eigenvalues of A and the dimension of each eigenspace.
b) (3) Find the minimal polynomial of A. Is A diagonalizable?
c) (4) Find a formula for A^k, $k \geq 1$.

4. Restriction and quotient. Let V be finite-dimensional, $T \in \mathcal{L}(V)$, and U a T-invariant subspace. Write $T|_U \in \mathcal{L}(U)$ and $T/U \in \mathcal{L}(V/U)$.

a) (3) Prove that T/U, $(T/U)(v + U) = Tv + U$, is well defined and linear.
b) (3) Prove that $m_{T|_U}$ and $m_{T/U}$ both divide m_T.
c) (3) Prove that m_T divides $m_{T|_U} \cdot m_{T/U}$.
d) (3) Deduce that if T is diagonalizable then so is $T|_U$. Then give T and U with $T|_U$ and T/U both diagonalizable but T not.

§4.2 Practice Midterm B (40 points, 50 minutes)

1. A dual map. Define $T\colon \mathcal{P}_2(\mathbb{R}) \to \mathbb{R}^2$ by $T(p) = \big(p(0),\, p'(1)\big)$.

a) (3) Find $\mathcal{M}(T)$ with respect to $1, x, x^2$ and the standard basis e_1, e_2.

b) (3) Let ψ_1, ψ_2 be the dual basis of e_1, e_2 and $\varphi_0, \varphi_1, \varphi_2$ the dual basis of $1, x, x^2$. Compute $T'(\psi_1)$ and $T'(\psi_2)$ in terms of the φ_j.

c) (2) Find $\operatorname{null} T'$, and a basis of $\operatorname{null} T$. Check that $\operatorname{range} T' = (\operatorname{null} T)^0$ has the right dimension.

2. Commuting operators. Let V be a nonzero finite-dimensional complex vector space and $S, T \in \mathcal{L}(V)$ with $ST = TS$.

a) (2) Prove that each eigenspace $E(\lambda, T)$ is S-invariant.

b) (3) Prove that S and T have a common eigenvector.

c) (3) Assume S and T are diagonalizable. Prove there is a basis of V of common eigenvectors. (You may quote: the restriction of a diagonalizable operator to an invariant subspace is diagonalizable.)

d) (2) Let $T = \operatorname{diag}(1, 2, 3)$ on $\mathbb{C}^3$. Prove that every S commuting with T is diagonal.

3. A quotient of matrices. Let $V = \mathbb{R}^{2,2}$ and $T\colon V \to \mathbb{R}^2$, $T(A) = A\left(\begin{smallmatrix}1\\-1\end{smallmatrix}\right)$. Let $N = \operatorname{null} T$.

a) (3) Prove T is linear and surjective, and find a basis of N.

b) (3) Write down the isomorphism $V/N \to \mathbb{R}^2$ given by the first isomorphism theorem, and find a basis of V/N.

c) (4) Fix $B \in \mathbb{R}^{2,2}$ and let $L \in \mathcal{L}(V)$, $L(A) = BA$. Prove N is L-invariant, and that under the isomorphism of (b), L/N corresponds to $x \mapsto Bx$ on $\mathbb{R}^2$. Find the eigenvalues of L/N when $B = \left(\begin{smallmatrix}2&1\\0&3\end{smallmatrix}\right)$.

4. Minimal polynomials. Let V be finite-dimensional and $T \in \mathcal{L}(V)$.

a) (4) Prove that $\lambda \in \mathbb{F}$ is an eigenvalue of T iff $m_T(\lambda) = 0$.

b) (4) Prove that $m_{T-\lambda I}(x) = m_T(x + \lambda)$ for every $\lambda \in \mathbb{F}$.

c) (4) $T \in \mathcal{L}(\mathbb{C}^4)$ satisfies $T^2 - 3T + 2I = 0$, and $T \neq I$, $T \neq 2I$. Prove T is diagonalizable, find m_T, and list all possible pairs $\big(\dim E(1, T), \dim E(2, T)\big)$.

§4.3 Practice Midterm C (40 points, 50 minutes)

1. Eigenvectors of a differential operator. Define $T\colon \mathcal{P}_2(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ by $(Tp)(x) = (x+1)p'(x)$.

a) (3) Find $\mathcal{M}(T)$ with respect to $1, x, x^2$, and the eigenvalues of T.
b) (3) Find an eigenbasis of T.
c) (2) Compute $T^k(x^2)$ for $k \geq 1$.

2. Products and sums. Let V, W be finite-dimensional, and U_1, U_2, U_3 subspaces of V.

a) (3) Prove $\dim(V \times W) = \dim V + \dim W$ by giving a basis of $V \times W$.
b) (3) Let $\Gamma\colon U_1 \times U_2 \to V$, $\Gamma(u_1, u_2) = u_1 + u_2$. Find $\operatorname{null}\Gamma$ and deduce $\dim(U_1 + U_2) = \dim U_1 + \dim U_2 - \dim(U_1 \cap U_2)$.
c) (4) Give three subspaces of $\mathbb{R}^2$ with $U_1 \cap U_2 = U_1 \cap U_3 = U_2 \cap U_3 = \{0\}$ whose sum is not direct. For your example, find a nonzero vector in the null space of $\Gamma\colon U_1 \times U_2 \times U_3 \to \mathbb{R}^2$, $\Gamma(u_1, u_2, u_3) = u_1 + u_2 + u_3$.

3. A minimal polynomial. Let

$$A = \begin{pmatrix} 0 & 0 & 2 \\ 1 & 0 & -1 \\ 0 & 1 & 2 \end{pmatrix}.$$

a) (3) Compute Ae_1, A^2e_1, A^3e_1. Deduce the minimal polynomial of A.
b) (3) Find the eigenvalues of A over $\mathbb{R}$ and over $\mathbb{C}$. Is A diagonalizable over $\mathbb{R}$? Over $\mathbb{C}$?
c) (4) Prove A is invertible and write A^{-1} as a polynomial in A.

4. Annihilators. Let V be finite-dimensional and U, W subspaces.

a) (3) Prove $\dim U^0 = \dim V - \dim U$, using the restriction map $V' \to U'$, $\varphi \mapsto \varphi|_U$. (You may use that it is surjective.)
b) (3) Prove $(U + W)^0 = U^0 \cap W^0$.
c) (3) Let $T \in \mathcal{L}(V, X)$. Prove $\dim\operatorname{null} T' = \dim\operatorname{null} T + \dim X - \dim V$.
d) (3) Let $\varphi \in V'$ be nonzero and $U = \operatorname{null}\varphi$. Prove $U^0 = \operatorname{span}(\varphi)$. Find a basis of U^0 for $U = \{p \in \mathcal{P}_2(\mathbb{R}) : p(1) = 0\}$.

§5 Problem bank

Problem 1 (Two roots (Ribet, Spring 2014))**.**

(a) $(T-aI)(T-bI)=0$ and T is not a scalar multiple of I. Prove that T is diagonalizable if and only if $a\neq b$.
(b) With the same hypotheses on $\mathbb{C}^n$ and $a\neq b$, prove $\mathbb{C}^n = E(a,T)\oplus E(b,T)$ and $1\leq \dim E(a,T)\leq n-1$.
(c) Now suppose only $(T-aI)^2=0$. Must T be diagonalizable? Determine exactly when it is.

Problem 2 (Eigenvalues of the dual map (Holtz, Fall 2015))**.**

(a) Prove $(T-\lambda I)' = T'-\lambda I$.
(b) Prove $E(\lambda,T') = \big(\operatorname{range}(T-\lambda I)\big)^0$.
(c) For $T(x,y)=(x+y,y)$ on $\mathbb{R}^2$, find $E(1,T')$ in terms of the dual basis φ_1,φ_2 of e_1,e_2, and check it against (b).
(d) Prove T is diagonalizable iff T' is.

Problem 3 (Cyclic vectors (Ribet, Spring 2020, extended))**.**

(a) Let $A=\left(\begin{smallmatrix}1&1&0\\0&1&0\\0&0&2\end{smallmatrix}\right)$ on $\mathbb{R}^3$. Find the smallest A-invariant subspace containing $v=(1,0,1)$ and the minimal polynomial of A restricted to it. Then show $(1,1,1)$ is a cyclic vector and deduce m_A.
(b) Suppose $T\in\mathcal{L}(V)$ has a cyclic vector v. Prove that every $S\in\mathcal{L}(V)$ with $ST=TS$ is a polynomial in T.
(c) Show that (b) fails for $T=I$ on $\mathbb{F}^2$.

Problem 4 (Diagonalizing a differential operator (Persson, Fall 2010))**.** Let $T\colon \mathcal{P}_3(\mathbb{R})\to\mathcal{P}_3(\mathbb{R})$, $Tp=(x^2+1)p''$.

(a) Find $\mathcal{M}(T)$ in the basis $1,x,x^2,x^3$ and the eigenvalues of T.
(b) Decide whether T is diagonalizable, and if so find an eigenbasis.
(c) Find the minimal polynomial of T.

Problem 5 (A quotient of $\mathcal{P}_3(\mathbb{R})$ (HW5 P1 style))**.** Let $U=\{p\in\mathcal{P}_3(\mathbb{R}) : p(1)=p(-1)=0\}$ and $W=\{p\in\mathcal{P}_3(\mathbb{R}) : p(1)=0\}$.

(a) Find a basis of U, an explicit isomorphism $\mathcal{P}_3(\mathbb{R})/U\to\mathbb{R}^2$, and a basis of $\mathcal{P}_3(\mathbb{R})/U$.
(b) Let D be differentiation. Is U invariant under D? Does $p+U\mapsto p'+U$ define a map $\mathcal{P}_3(\mathbb{R})/U\to\mathcal{P}_3(\mathbb{R})/U$?
(c) Show $U\subseteq W$, and use the third isomorphism theorem to identify $(\mathcal{P}_3(\mathbb{R})/U)/(W/U)$ with a familiar space.

Problem 6 (Eigenbasis from a null space test (Ribet, Spring 2020 final))**.** Let V be a finite-dimensional complex vector space.

(a) Prove: if $\operatorname{null}(T-\alpha I)^2=\operatorname{null}(T-\alpha I)$ for every $\alpha\in\mathbb{C}$, then T is diagonalizable.
(b) Prove the converse.
(c) Show that in (a) it is enough to check the eigenvalues α of T, and give a non-diagonalizable T and an α for which $\operatorname{null}(T-\alpha I)^2\neq\operatorname{null}(T-\alpha I)$.

Problem 7 (Square roots and diagonalizability)**.** Let V be a finite-dimensional complex vector space.

(a) Prove: if T is invertible and T^2 is diagonalizable, then T is diagonalizable.
(b) Show that "invertible" cannot be dropped.
(c) Prove: if T is diagonalizable, then $T=S^2$ for some diagonalizable S.
(d) Prove that $\left(\begin{smallmatrix}0&1\\0&0\end{smallmatrix}\right)$ has no square root (no S with S^2 equal to it).

Problem 8 (Commuting operators (LADR 5.80, 5.81))**.** Let V be a nonzero finite-dimensional complex vector space and $S,T\in\mathcal{L}(V)$ with $ST=TS$.

(a) Prove there is one basis in which $\mathcal{M}(S)$ and $\mathcal{M}(T)$ are both upper-triangular.
(b) Prove every eigenvalue of $S+T$ has the form $\alpha+\beta$, and every eigenvalue of ST the form $\alpha\beta$, with α an eigenvalue of S and β an eigenvalue of T.
(c) Show that (b) fails for $S=\left(\begin{smallmatrix}0&1\\0&0\end{smallmatrix}\right)$, $T=\left(\begin{smallmatrix}0&0\\1&0\end{smallmatrix}\right)$.

Problem 9 (Using the isomorphism theorems)**.**

(a) In $\mathcal{P}_3(\mathbb{R})$ let $U=\{p : p(0)=0\}$ and $W=\{p : p(1)=0\}$. Find $U\cap W$ and $U+W$, write down the isomorphism $U/(U\cap W)\to(U+W)/W$ of the second isomorphism theorem, and identify both spaces with $\mathbb{R}$.
(b) Prove: if $V=U\oplus W$, then $V/U\cong W$.
(c) Let $T\in\mathcal{L}(V,W)$ and let U be a subspace of V. Prove $T(U)\cong U/(U\cap\operatorname{null}T)$.

Problem 10 (Functionals separating vectors (Ribet, Spring 2010, Spring 2014))**.** Let V be finite-dimensional.

(a) If $v_1, \dots, v_m$ are linearly independent, prove there are $\varphi_1, \dots, \varphi_m \in V'$ with $\varphi_i(v_j) = 1$ if $i = j$ and 0 otherwise.
(b) Prove $W = \{v \in V : \varphi(v) = 0 \text{ for all } \varphi \in W^0\}$.
(c) Prove $U \subseteq W$ iff $W^0 \subseteq U^0$.

Problem 11 (Counting invariant subspaces)**.**

(a) Find all invariant subspaces of $T = \operatorname{diag}(1, 2, 3)$ on $\mathbb{C}^3$.
(b) Find all invariant subspaces of N on $\mathbb{C}^3$, where $Ne_1 = 0$, $Ne_2 = e_1$, $Ne_3 = e_2$.
(c) Find all invariant subspaces of I on $\mathbb{C}^2$. How many are there?

Problem 12 (Polynomial identities)**.**

(a) $T \in \mathcal{L}(\mathbb{C}^3)$ satisfies $T^2 = 2T - I$. Prove that 1 is the only eigenvalue of T, and that T is diagonalizable iff $T = I$.
(b) (Ribet, Spring 2005) If $(A - 2I)(A - 3I)(A - 4I)(A - 5I) = 0$ for a real $n \times n$ matrix A with $n \geq 1$, prove at least one of $2, 3, 4, 5$ is an eigenvalue, and that A is diagonalizable.
(c) T is invertible with minimal polynomial $x^3 - 2x + 5$. Write T^{-1} as a polynomial in T, and find the minimal polynomial of T^{-1}.

§6 True or false

Give a one-line reason or counterexample. Spaces are finite-dimensional unless stated otherwise.

1. $V \times W$ and $W \times V$ are isomorphic.
2. $\dim(V \times W) = (\dim V)(\dim W)$.
3. If $U_1 \cap U_2 = U_1 \cap U_3 = U_2 \cap U_3 = \{0\}$, then $U_1 + U_2 + U_3$ is a direct sum.
4. An affine subset $v + U$ contains 0 iff $v \in U$.
5. If $v + U = w + U$, then $v = w$.
6. $V/\{0\} \cong V$ and $V/V = \{0 + V\}$.
7. If $T\colon V \to W$ is surjective, then $V/\operatorname{null} T \cong W$.
8. If $V/U \cong V/W$, then $U = W$.
9. $\dim U/(U \cap W) = \dim(U + W) - \dim W$.
10. Every linear map $V/U \to W$ has the form $v + U \mapsto Tv$ for some $T \in \mathcal{L}(V, W)$ with $U \subseteq \operatorname{null} T$.
11. If $U \subseteq \operatorname{null} T$ for $T \in \mathcal{L}(V, W)$, then $v + U \mapsto Tv$ is a well-defined linear map $V/U \to W$.
12. If $\varphi \in V'$ is nonzero, then $\dim \operatorname{null} \varphi = \dim V - 1$.
13. If $\varphi_1, \varphi_2 \in V'$ have the same null space, then $\varphi_1 = c\varphi_2$ for some scalar c.
14. If T is injective, then T' is injective.
15. $(U \cap W)^0 = U^0 \cap W^0$.
16. If $v_1, \dots, v_n$ is a basis with dual basis $\varphi_1, \dots, \varphi_n$, then φ_1 depends only on v_1.
17. Every operator on a nonzero real vector space has an eigenvalue.
18. If $T^2 = T$, then 0 and 1 are both eigenvalues of T.
19. The sum of two eigenvectors of T is an eigenvector of T or 0.
20. If v_1, v_2 and $v_1 + v_2$ are eigenvectors of T, then all three have the same eigenvalue. (Ribet, Spring 2014)
21. If λ is an eigenvalue of T, then λ^2 is an eigenvalue of T^2.
22. If A and B have the same eigenvalues, they are similar.
23. If U is T-invariant and T is invertible, then $T|_U$ is invertible.
24. A 2×2 matrix whose characteristic polynomial has a double root is not diagonalizable.
25. An upper-triangular matrix with distinct diagonal entries is diagonalizable. (Ribet, Spring 2005)
26. If T is invertible and diagonalizable, then T^{-1} is diagonalizable.
27. If T is diagonalizable and U is T-invariant, then $T|_U$ is diagonalizable.
28. If S and T are diagonalizable, so is $S + T$.
29. If $T^2 = I$ on a real space, then T is diagonalizable.
30. If $T^2 = 0$, then T is diagonalizable.
31. Every operator on a real space has an upper-triangular matrix in some basis.
32. $\deg m_T \leq \dim V$.
33. If $p(T) = 0$, then every root of p is an eigenvalue of T.

34. If T is invertible, then $m_T(0) \neq 0$.
35. If $m_T = (x-1)(x-2)$, then $\dim V = 2$.
36. If T has $\dim V$ distinct eigenvalues, then T has a cyclic vector.
37. If A is a real 5×5 matrix, then A has a real eigenvector. (Persson, Fall 2010)